

SequAna Genomic Course

QC on Genome Assembly & Genome Annotation

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The Vertebrate Genomes Project



We aim to generate near **error-free reference genome assemblies** and **genome annotations** of all 70,000 extant vertebrate species.



All data store in a publicly available in an open-access library of genomes (GenomeArk).



vgp.github.io/genomeark/

GenomeArk

Vertebrate Genomes Project



Acanthisitta chloris

rifleman

some data | no assembly



Acipenser ruthenus

sterlet

all data | high-quality draft assembly



Alca torda

razorbill

all data | curated assembly | mito assembly



Amblyraja radiata

thorny skate

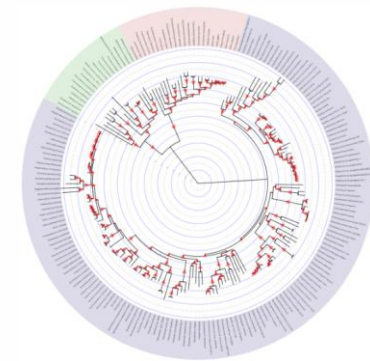
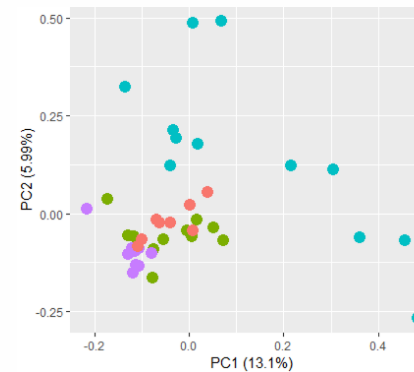
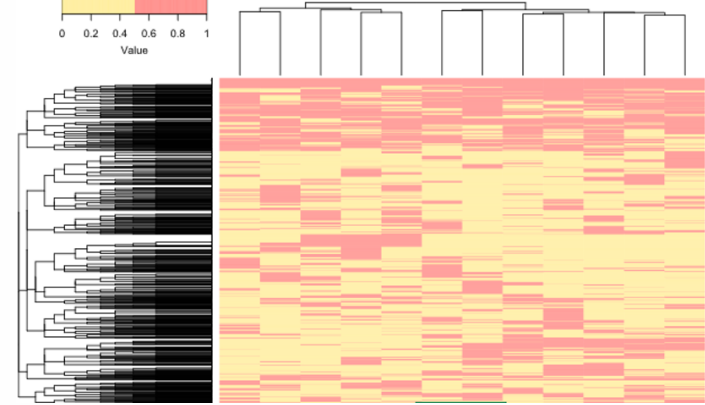
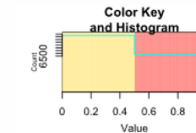
all data | high-quality draft assembly | mito assembly

The Vertebrate Genomes Project



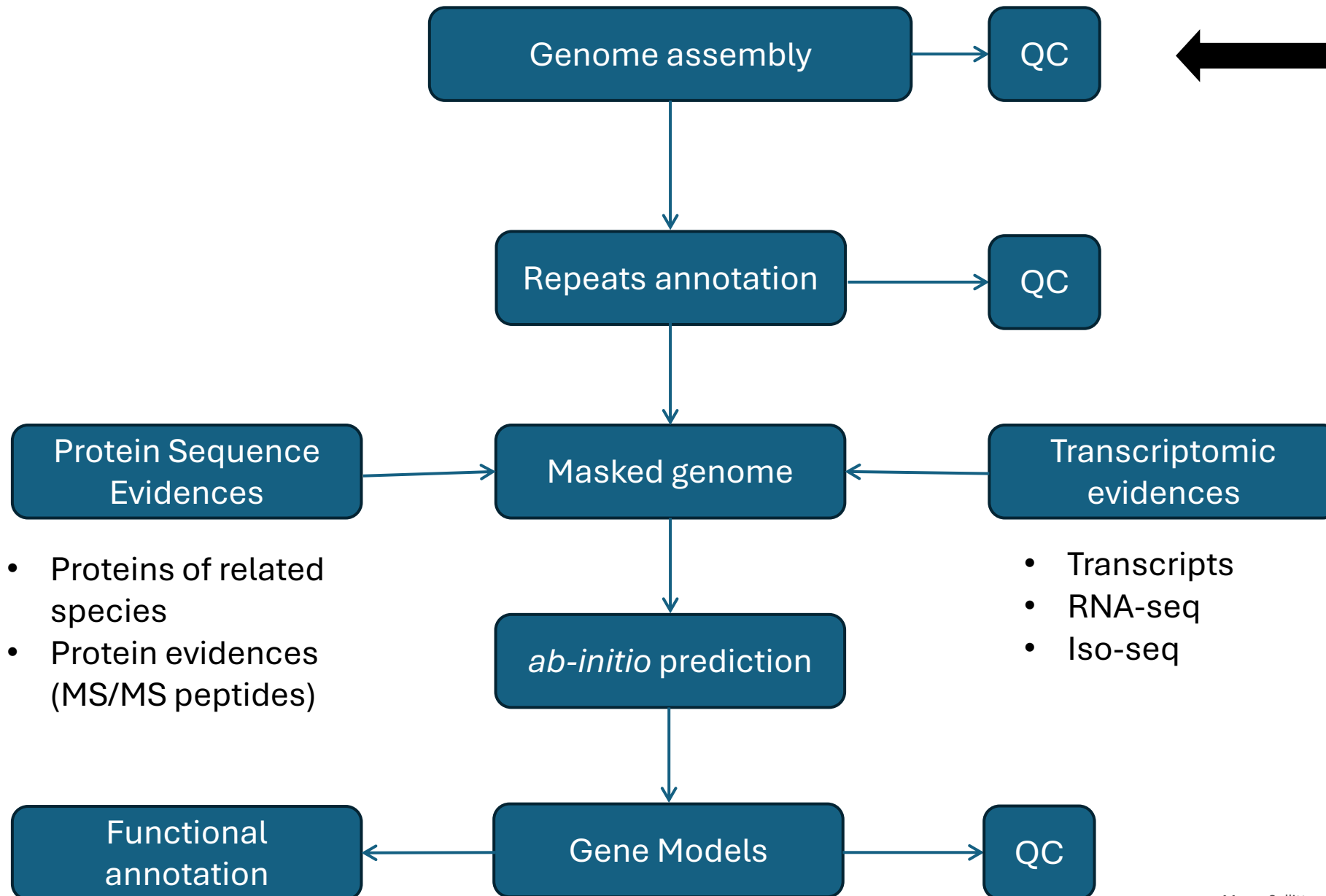
Main goals:

- Phylogeny
- Chromosome evolution
- Comparative genomics of traits and disease
- Disease resistance genes
- Conservation genomics



Outline of the seminar

- QC on Genome Assembly
- Overview on TE annotation
- Introduction on the annotation process
- Pipelines for the genome annotation
- Focus on EGAP/EGAPx Pipeline
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Contiguity: N50/L50

- ❖ **N50:** The sequence length of the shortest contig at 50% of the total genome/assembly length
- ❖ **L50:** L50 is the smallest number of sequences whose combined length makes up at least 50% of the total genome assembly size.



- 1) Order the sequences according to their length
- 2) Sum sequences' lengths
- 3) Calculate the 50% of the total length
- 4) Identify the N50 and the L50

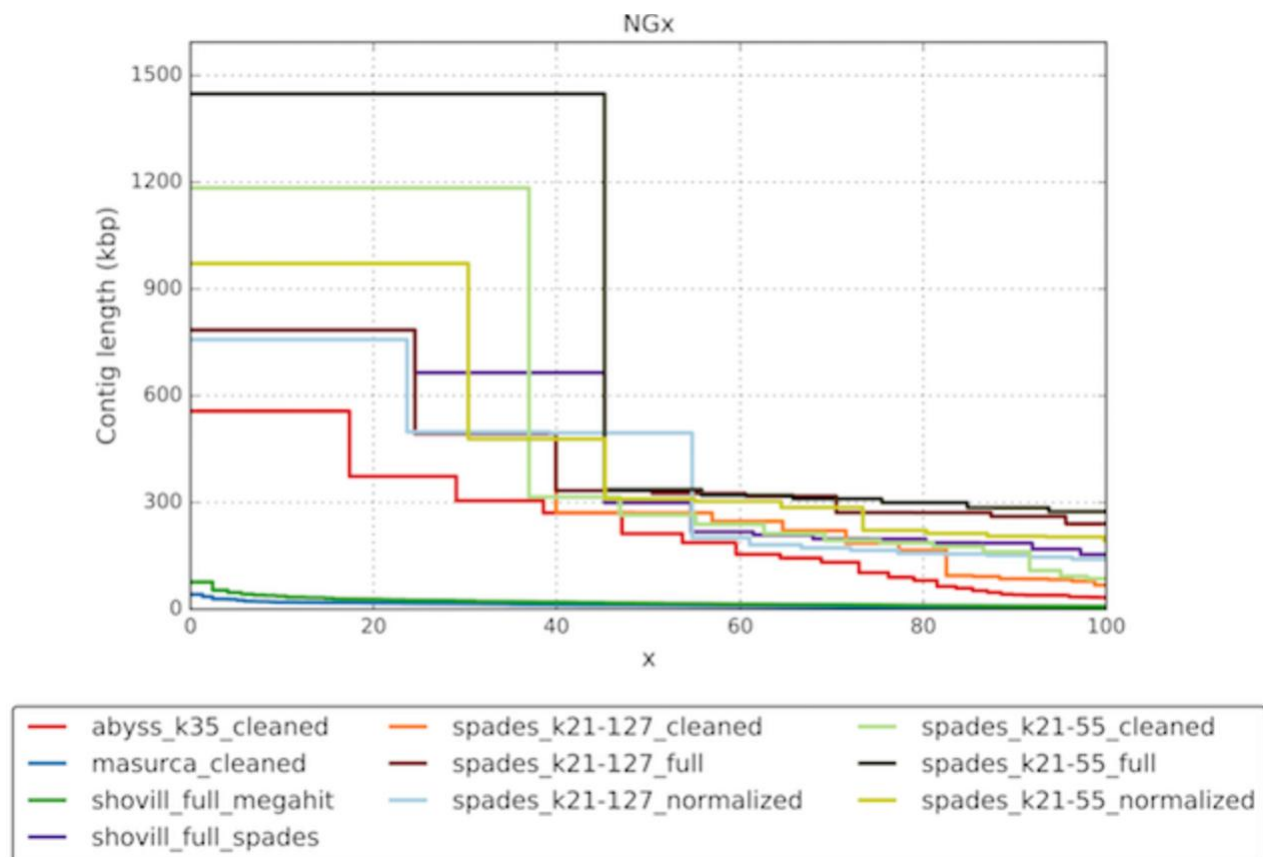
N50: Potential limitations ?

- Contiguity is different from completeness. Some content can be still missing
- Different assemblies can have the same N50, but in one case there may still be a high proportion of small contigs.
- N50 considers only the total length of the assembled contigs or scaffolds. In contrast, NG50 is more informative, as it is calculated relative to the estimated genome size.

Contiguity: auN

“auN”, as a measurement of
contiguity

$$\text{auN} = \sum_i L_i \cdot \frac{L_i}{\sum_j L_j} = \sum_i L_i^2 / \sum_j L_j$$

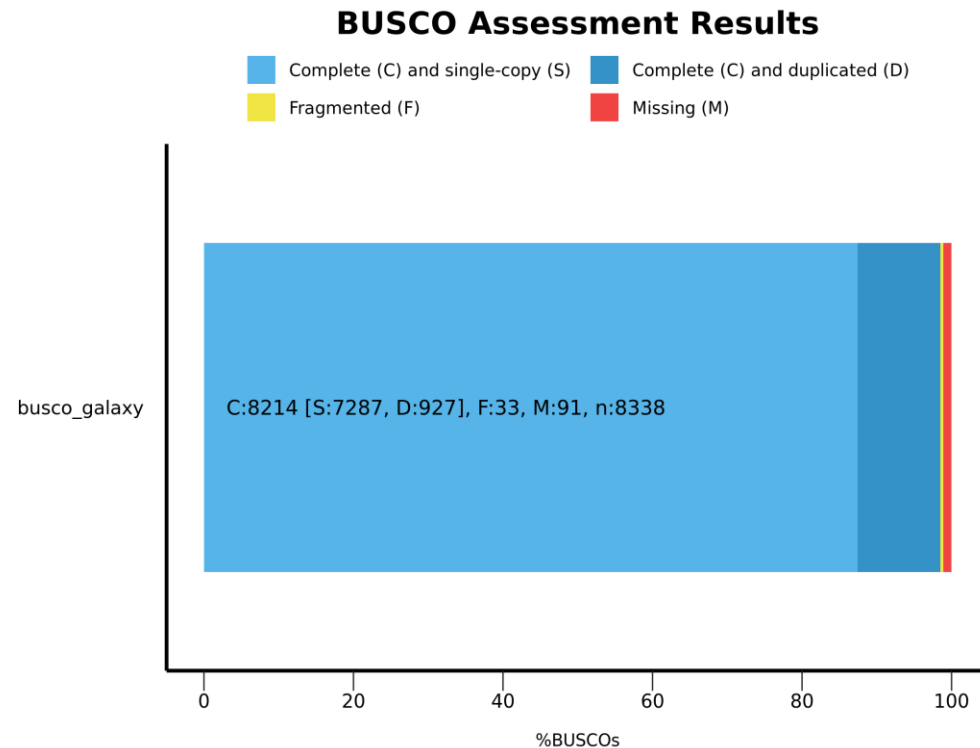
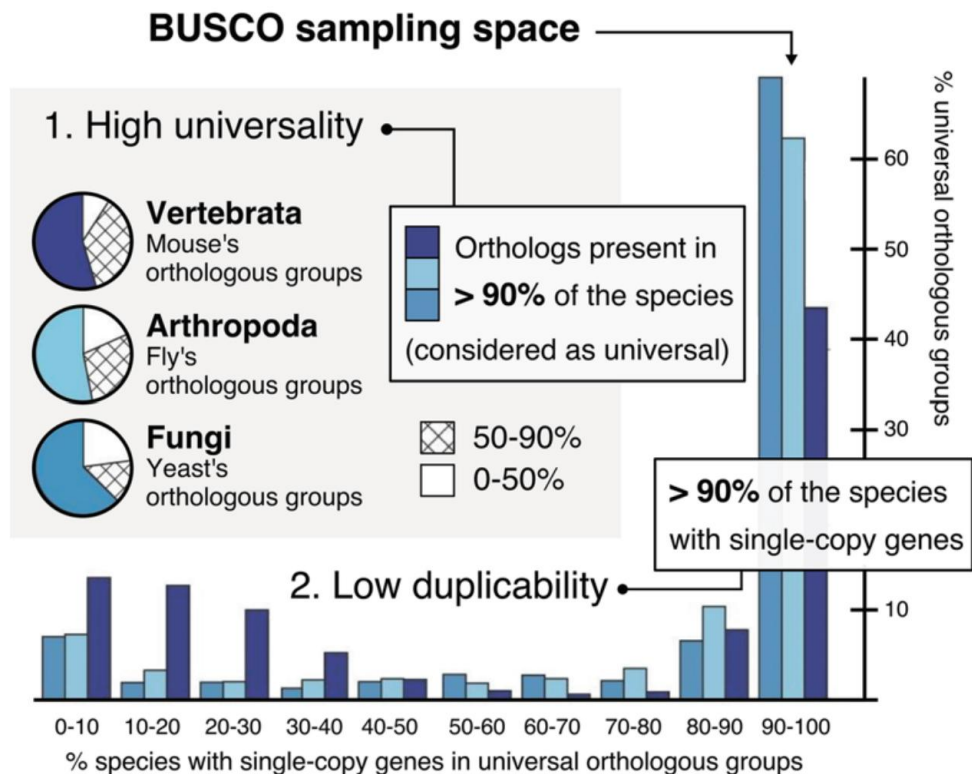


Tools for contiguity:
QUAST, gfastats,
assembly-stats

<https://lh3.github.io/2020/04/08/a-new-metric-on-assembly-contiguity>

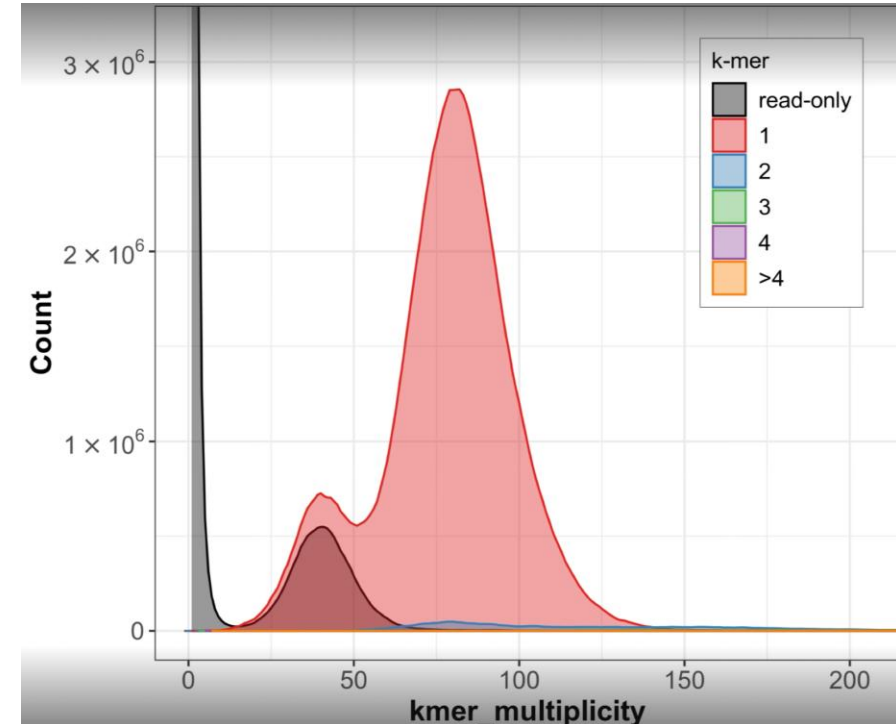
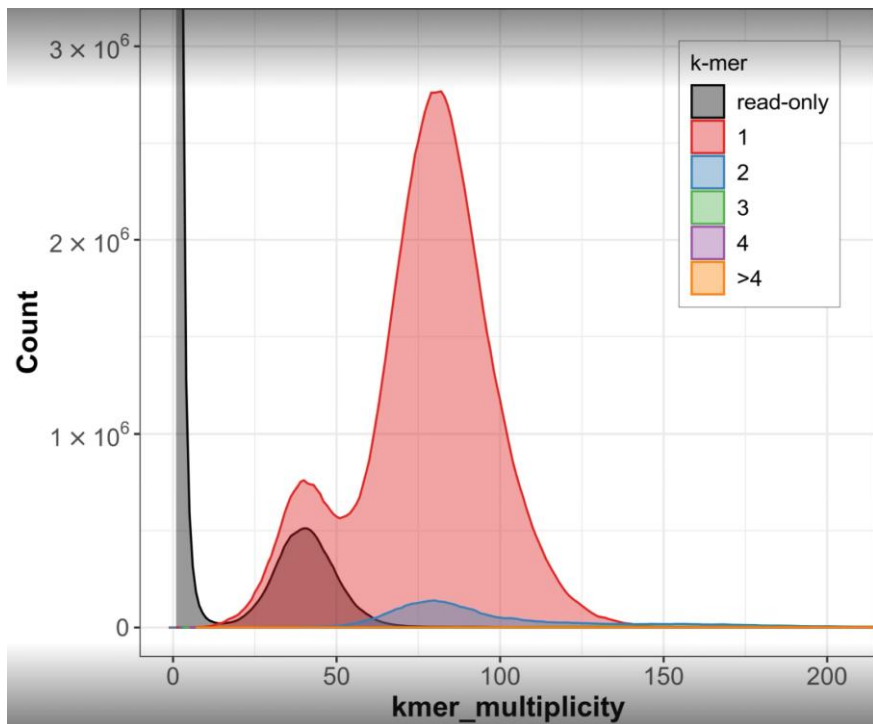
Completeness: BUSCO/Compleasm

- <https://busco.ezlab.org/> (Galaxy, Conda package and Docker container)
- Evaluates the presence of highly conserved single-copy orthologs expected in a given lineage, reporting the fraction of genes that are single-copy, duplicated, fragmented or missing



Completeness: kmer evaluation with Merqury/KAT

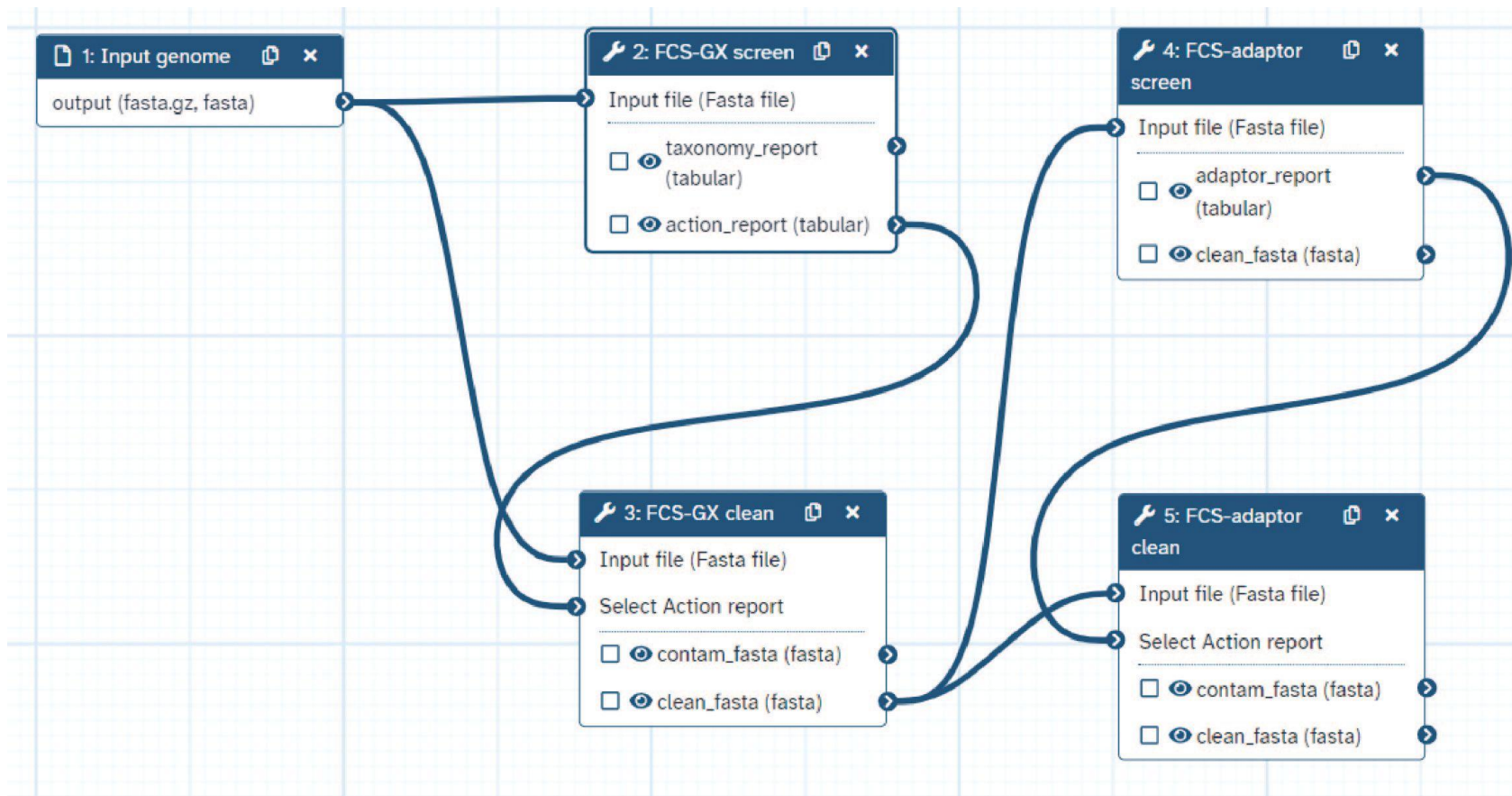
- Choose the “k”. Higher k increases unique k-mers but also error sensitivity; lower k reduces errors but increases repetitive k-mers.
- Evaluates k-mers found in the sequencing data with those present in the assembly.
- If most reliable read k-mers are also found in the assembly, sequence completeness is high.
- Estimates both **completeness** and **consensus quality**. A higher QV means fewer base-level errors in the assembly.
- In heterozygous diploids, two peaks usually appear: 1-copy k-mers = heterozygous ; 2-copy k-mers = homozygous
- Detect assembly problems -> reveal missing sequence, false duplications, and abnormal copy-number patterns.



Decontamination of the assembly: FCS

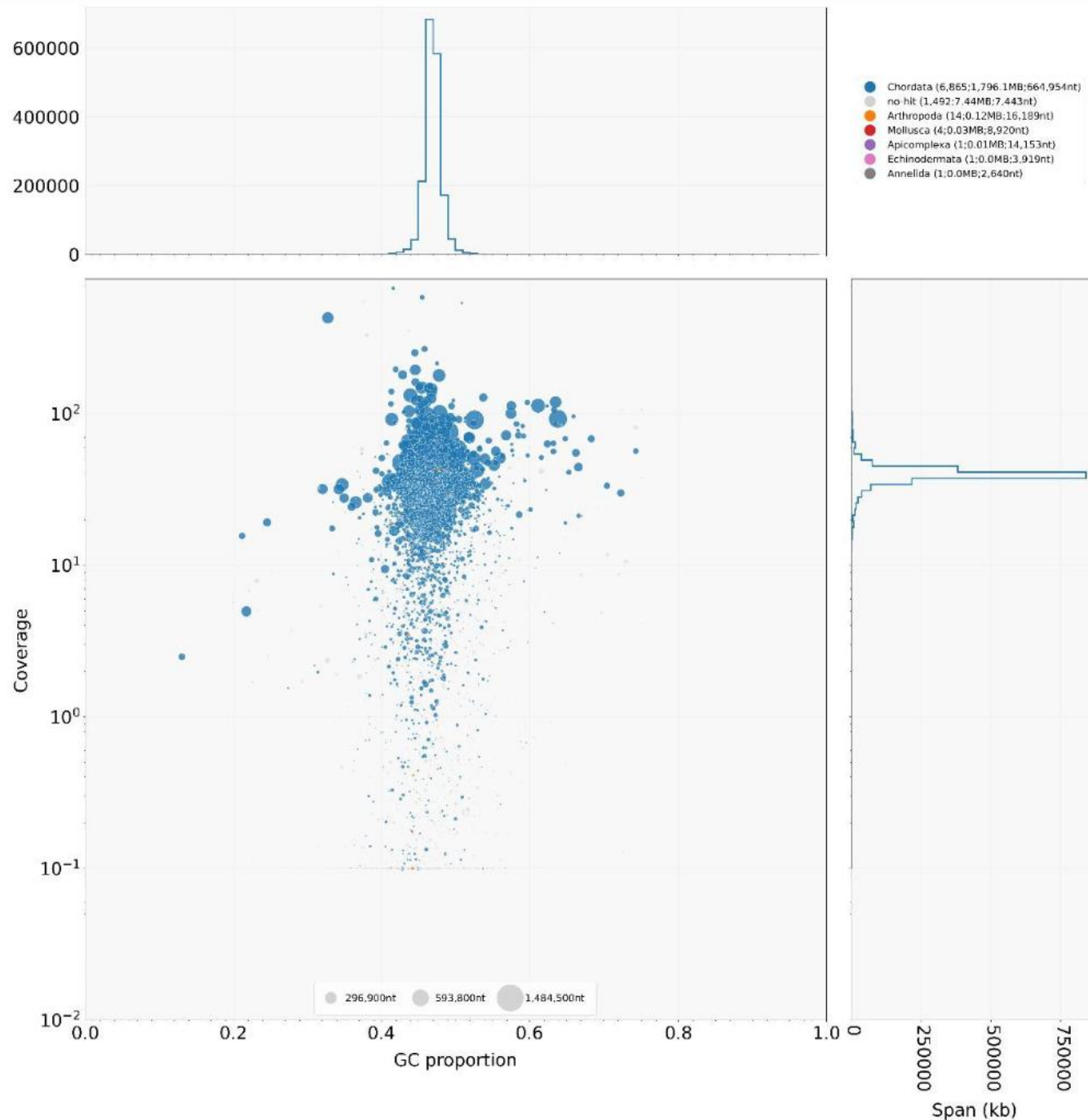
❖ Contamination:

Sequences in a dataset that do not biologically originate from the target organism. Contamination can be introduced at multiple stages of a genome assembly project.



- **FCS-adaptor** for contaminants from sequencing adaptors
- **FCS-GX** for contaminants from non-target organisms (Astashyn and Tvedte et al. Genome Biol 2024)

Decontamination of the assembly: Blobtools



Consider different features:

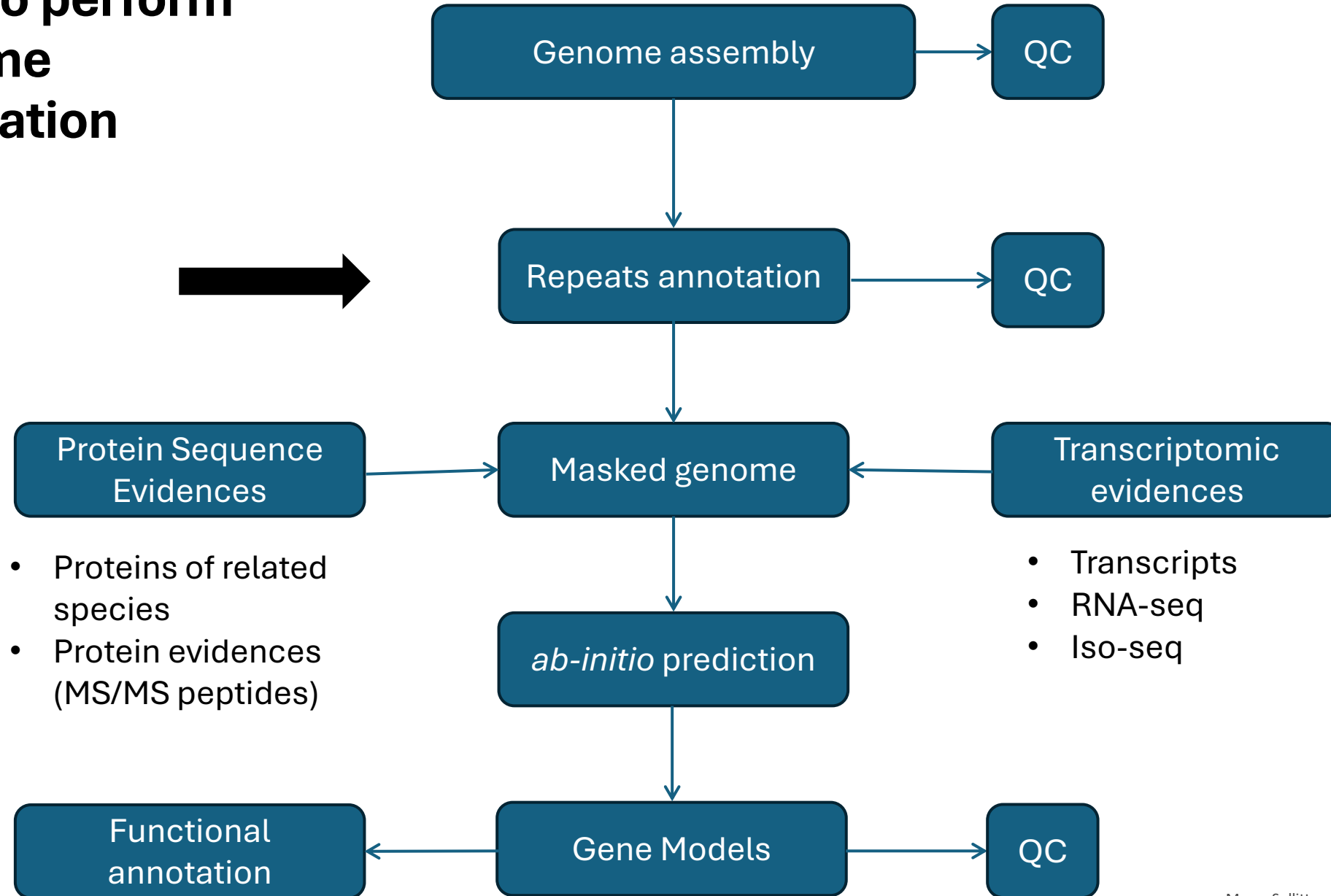
- 1) GC content
- 2) Coverage
- 3) Size of the sequences
- 4) Blast result of each sequence

<https://blobtoolkit.genomehubs.org/blobtools2>

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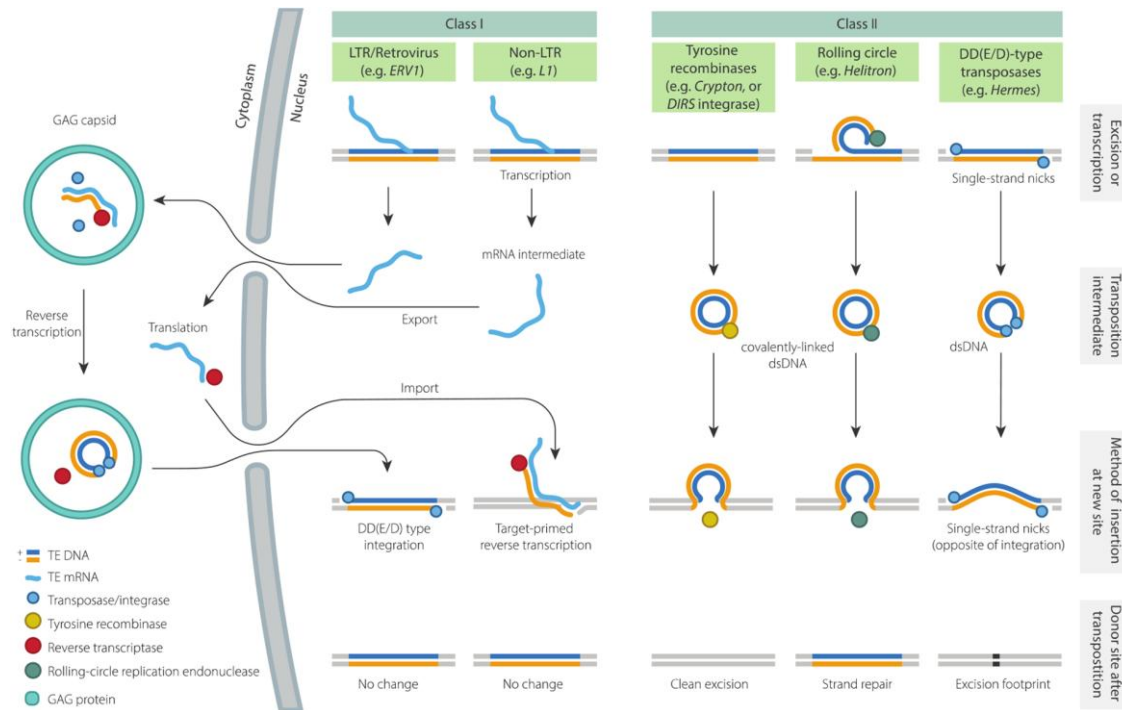
How to perform genome annotation



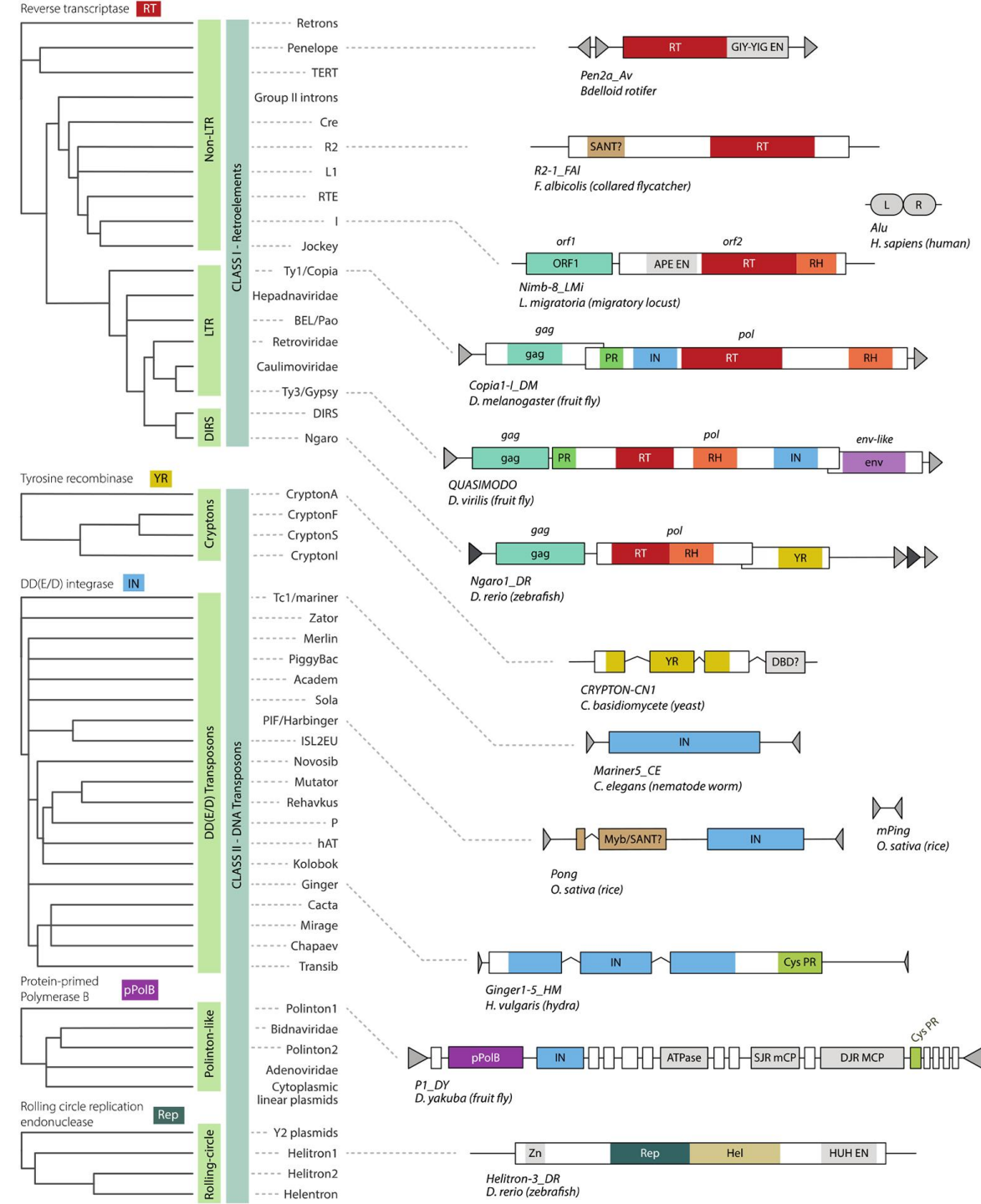
Repetitive elements

Repetitive DNA sequences (**repeats**) are nucleotide patterns that occur in multiple copies across the genome.

1. Satellite DNA
2. rDNA genes
3. Tandem or dispersed genes
4. tRNA genes
5. Transposable elements



Wells et al., 2020



Software used for the TE annotation and classification

Case 1 -> RepeatModeler2 + RepeatMasker
RepeatModeler2 -> includes RECON, RepeatScout, and an LTR identification pipeline based on LTRharvest/LTR_retriever. RepeatModeler2 creates a library of repeats that RepeatMasker uses to annotate and to mask the genome

softmasking: repeat bases are written in lower-cases ;
hardmasking: repeat bases are replaced by X

Pipelines to use

1. EarlGrey
2. EDTA2

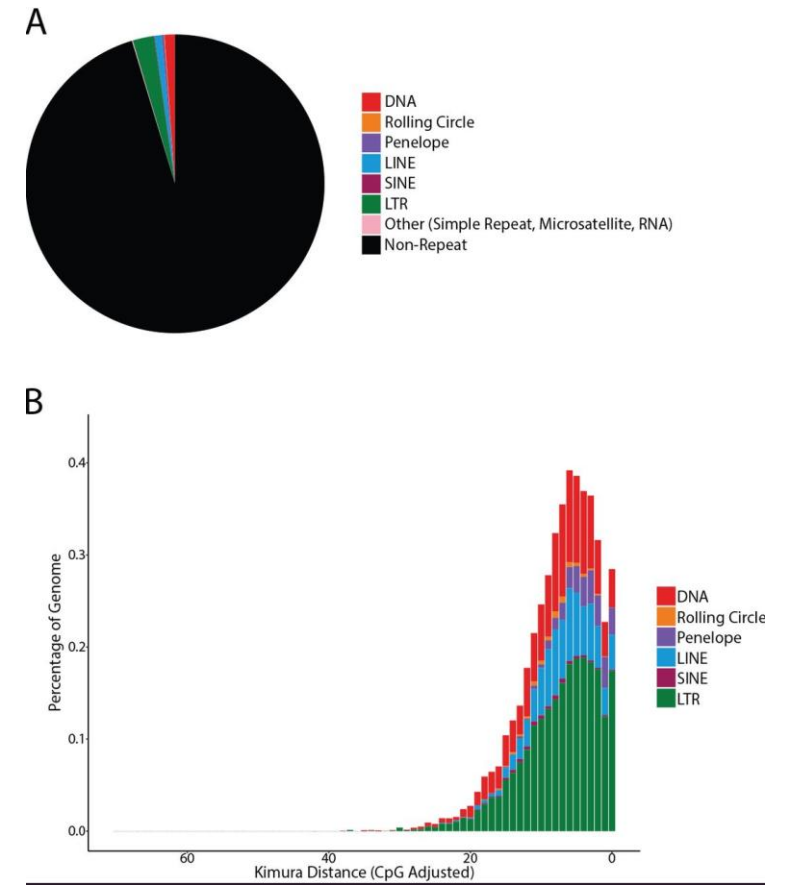
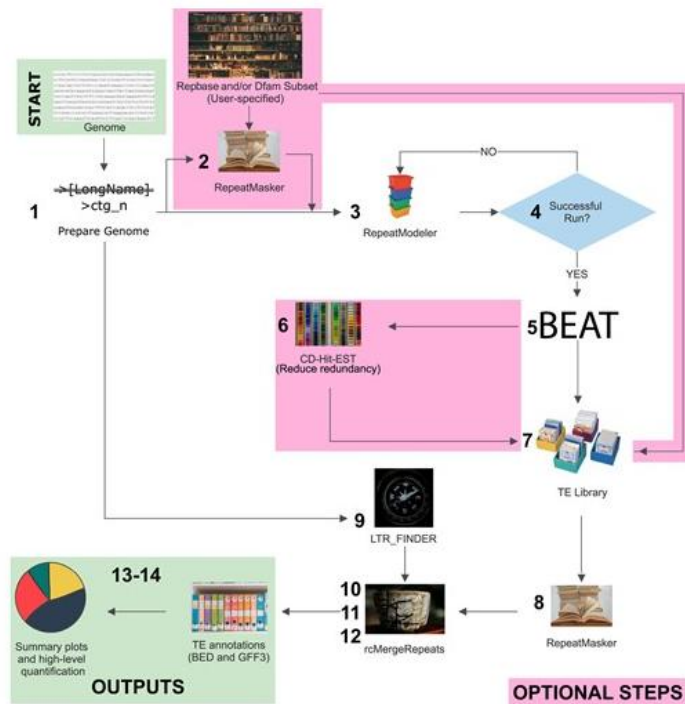
Software	Year	Type	Method	Development status	Presentation
CENSOR	1996	Annotator	Similarity	Maintained*	Web/Downloadable
ClassifyTE	2021	Classifier	Machine learning	Maintened	Downloadable
DAWGPAWS	2009	Annotator	Combined	Not maintained	Downloadable
DeepTE	2020	Classifier	Machine learning	Maintained	Downloadable
EarlGrey	2022	Annotator	Combined	Maintained	Downloadable
EDTA	2019	Annotator	<i>De novo</i>	Maintened	Downloadable
LTR annotator	2015	Annotator	<i>De novo</i>	Not maintained	Downloadable
LTR classifier	2016	Classifier	Library-based	Maintained	Web
LTR_finder	2007	Annotator	Structure	Not maintained	Downloadable
MITE-hunter	2010	Annotator	Structure	Not maintained	Downloadable
MITE-tracker	2018	Annotator	Structure	Maintained	Downloadable
PASTEC	2014	Classifier	Library-based	Maintained	Downloadable
reasonaTE	2022	Annotator	Combined	Maintained	Downloadable
REPCLASS	2015	Classifier	Library-based	Not maintained	Downloadable
RepeatClassifier	2020	Classifier	Library-based	Maintaned	Downloadable
RepeatMasker	1997	Annotator	Similarity	Maintained	Web/Downloadable
RepeatModeler	2008/2020	Annotator	<i>De novo</i>	Maintained	Downloadable
REPET	2011	Annotator	Combined	Maintained	Downloadable
RFSB	2022	Classifier	Machine learning	Maintained	Downloadable
RFSB	2022	Classifier	Machine learning	Maintaned	Downloadable
RTclass1	2010	Classifier	Library-based	Maintained	Web/Downloadable
TERL	2020	Classifier	Machine learning	Maintained	Downloadable
TEsorter	2022	Classifier	Library-based	Maintained	Downloadable
TIR-learner	2019	Annotator	Combined	Maintained	Downloadable

Earl Grey: A Fully Automated User-Friendly Transposable Element Annotation and Analysis Pipeline

Tobias Baril , James Galbraith, Alex Hayward  [Author Notes](#)

Molecular Biology and Evolution, Volume 41, Issue 4, April 2024, msae068,
<https://doi.org/10.1093/molbev/msae068>

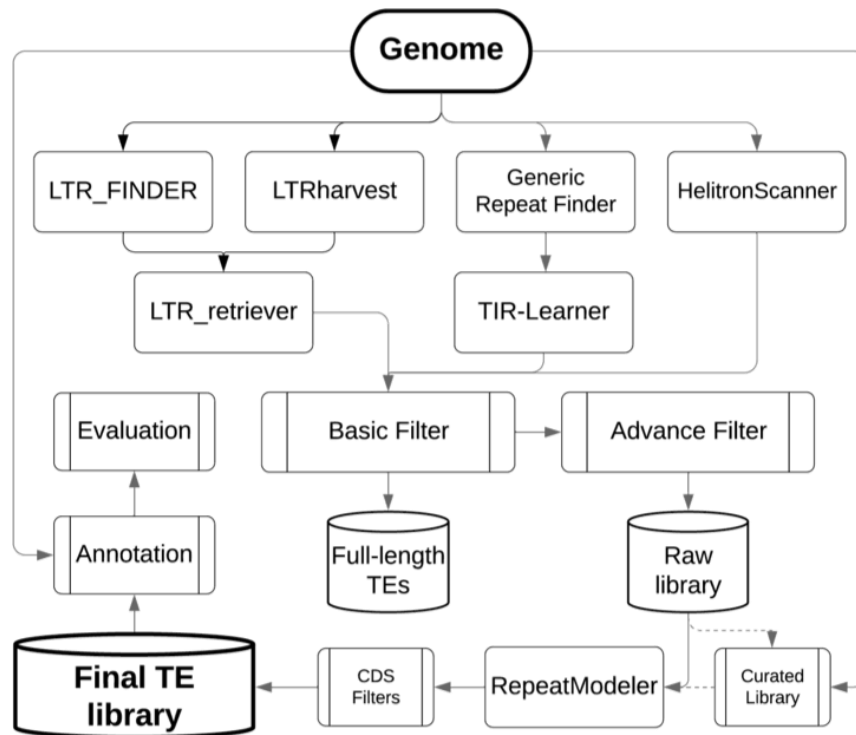
Published: 05 April 2024 **Article history** ▼



Benchmarking transposable element annotation methods for creation of a streamlined, comprehensive pipeline

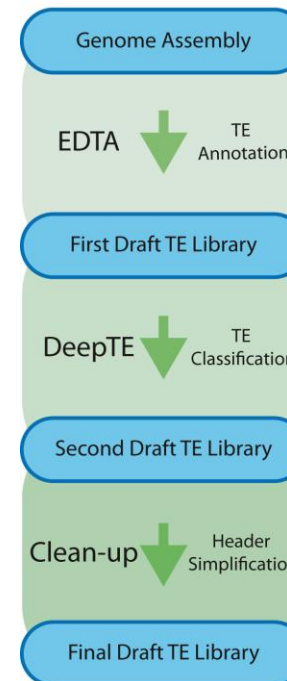
Research | [Open access](#) | Published: 16 December 2019

Volume 20, article number 275, (2019) [Cite this article](#)

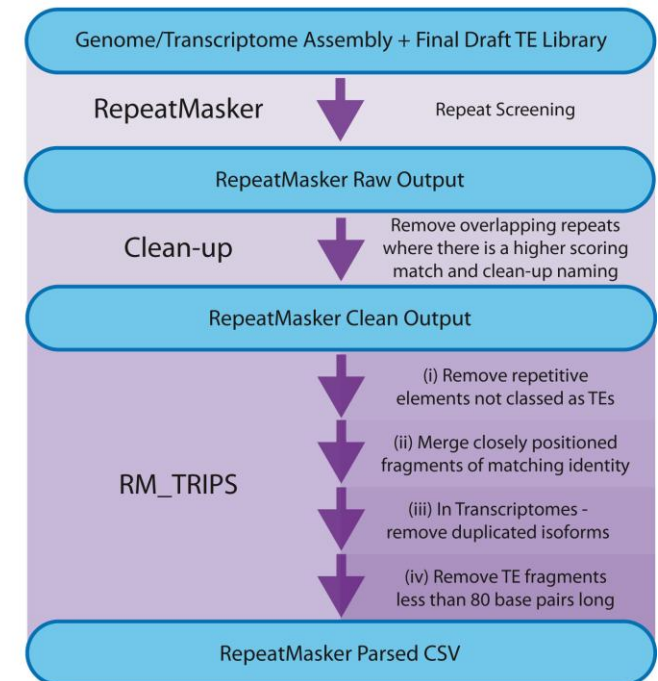


<https://github.com/oushujun/EDTA>

Part 1 TE Library Generation



Part 2 Screening for TEs



Bell et al., 2021

QC on repeats annotation

❖ Input

- TE annotations (GFF3/BED) from ≥ 2 workflows (e.g., EDTA, EarlGrey, RepeatModeler2 + RepeatMasker)
- TE reference/library (consensus)

❖ Basic QC

- % genome masked (overall and by TE class)
- Length & copy number distributions per TE family
- TE landscape by chromosome vs GC/content

❖ False Positive

- mark TE-derived gene models (sequence homology and protein identification)

❖ False Negatives

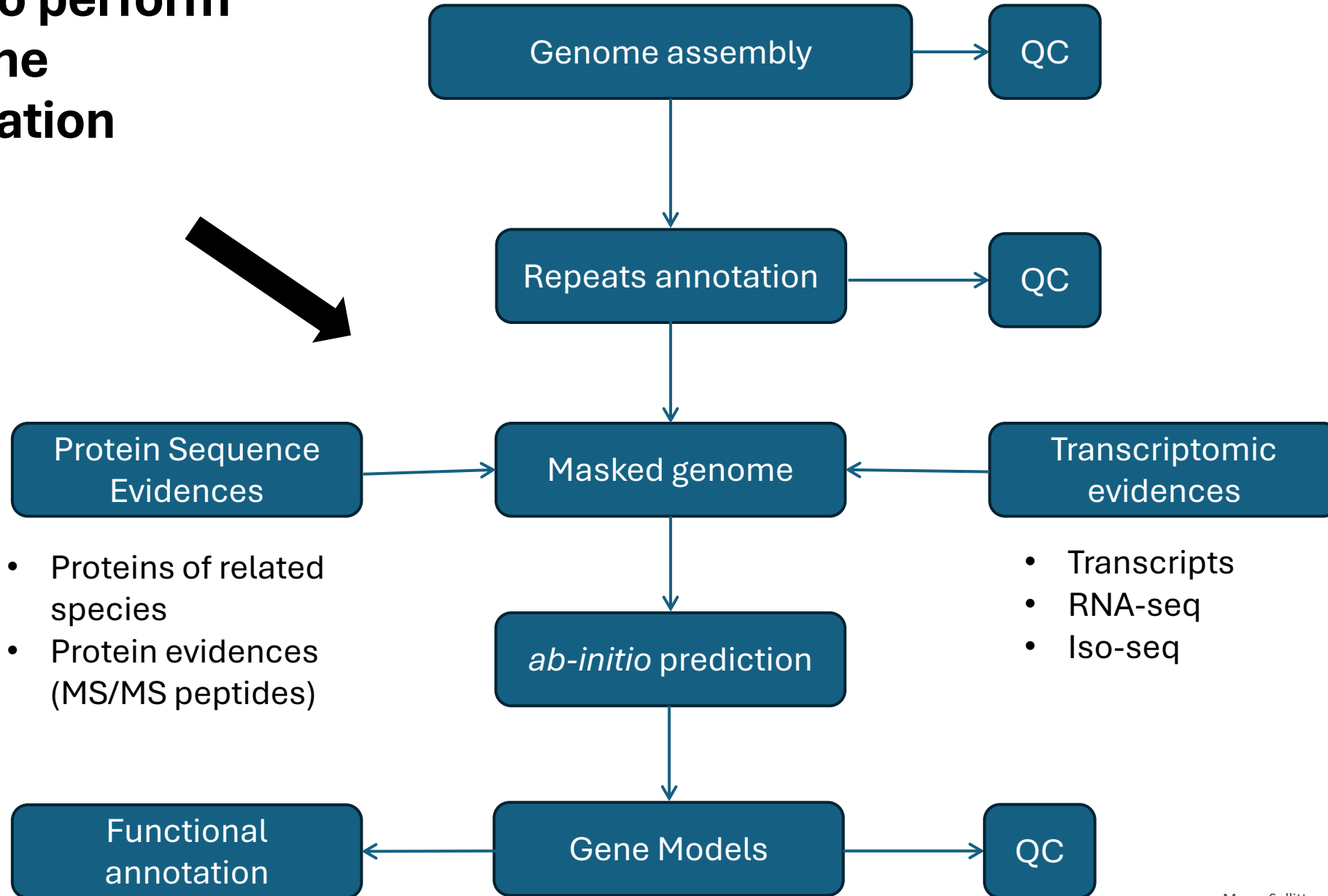
- Compare different tools' output (GFFs) using Bedtools ->
☐ **Jaccard index (A,B)** = $\text{Length (Intersection)} / (\text{Length (Union)} - \text{Length (Intersection)})$
- Hard-mask and rerun the masking process
- Kmer profiling (identification of kmers with high copy -> map back to the genome)

❖ Visualize tracks on IGV/Jbrowse2

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How to perform genome annotation



Transfer annotations using LiftOn/Liftoff

- Genome annotation lift-over tools -> To map genes between two assemblies of the same species or of closely related species
- LiftOn, homology-based software tool that integrates DNA and protein alignments to enhance the accuracy of genome-scale annotation and to allow mapping between relatively distant species.
- Liftoff transfers genes from a reference genome to a target genome by aligning sequences in a way that maximizes identity, while maintaining the structure of exons, transcripts, and genes.

Combining DNA and protein alignments to improve genome annotation with LiftOn

Kuan-Hao Chao ^{1 2}, Jakob M Heinz ³, Celine Hoh ^{1 2}, Alan Mao ^{1 2 4}, Alaina Shumate ^{2 4}, Mihaela Pertea ^{1 2 4}, Steven L Salzberg ^{1 2 4 5}

Affiliations + expand

PMID: 38798552 PMCID: [PMC11118573](#) DOI: [10.1101/2024.05.16.593026](#) [🔗](#)

JOURNAL ARTICLE

Liftoff: accurate mapping of gene annotations

FREE

Alaina Shumate [✉](#), Steven L Salzberg

Bioinformatics, Volume 37, Issue 12, June 2021, Pages 1639–1643,

<https://doi.org/10.1093/bioinformatics/btaa1016>

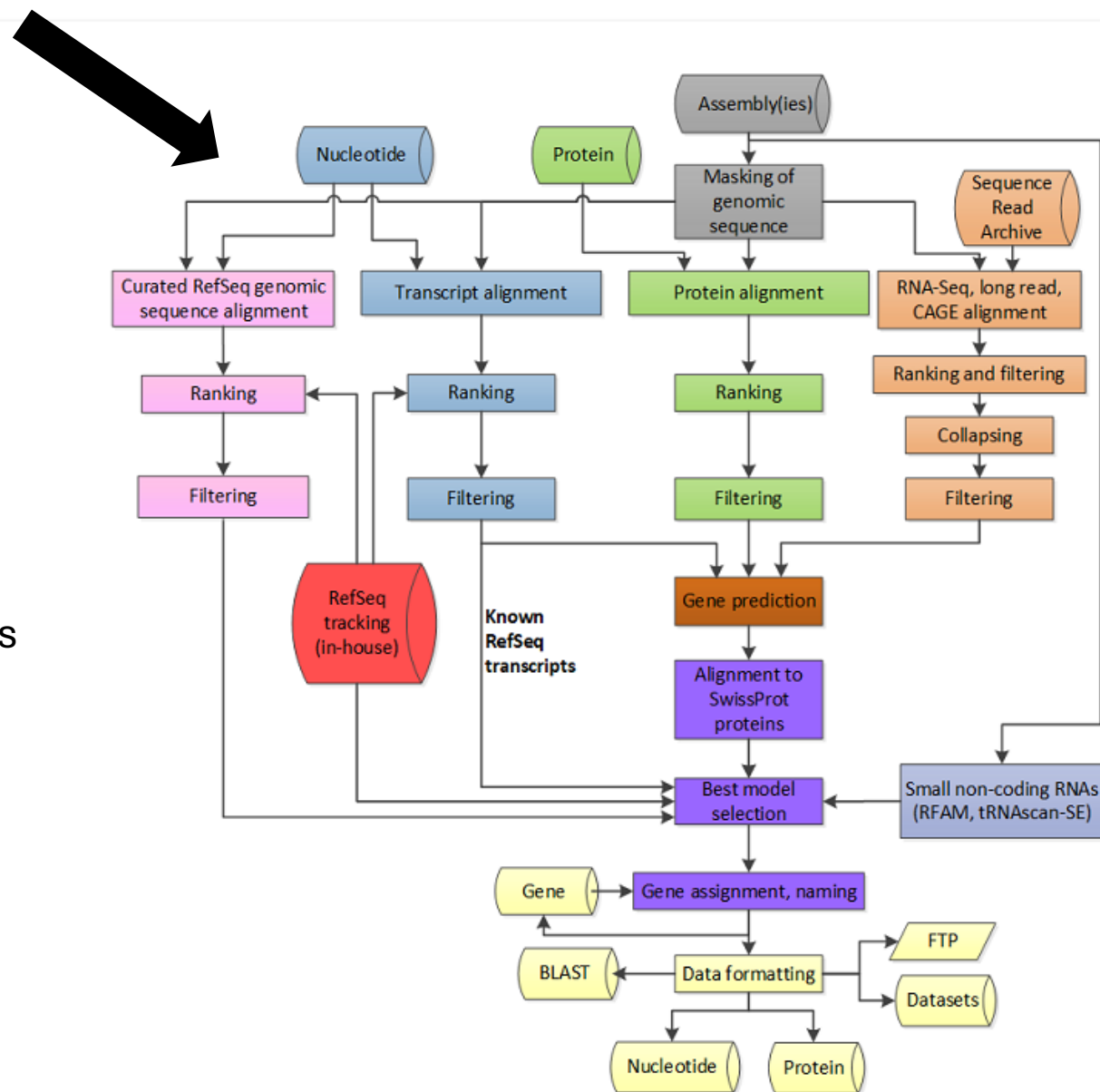
Published: 09 May 2021 **Article history** ▼

Pipelines/Tools for the *de novo* gene annotation

- 1) **The NCBI Eukaryotic Genome Annotation Pipeline (EGAP)** with RefSeq curation (https://www.ncbi.nlm.nih.gov/refseq/annotation_euk/process/) -> GNOMON
- 2) **EGAPx**, external pipeline based on EGAP (<https://github.com/ncbi/egapx>) -> GNOMON
- 3) EBI pipeline (ENSEMBL vertebrate and non-vertebrate pipeline) (<https://www.ebi.ac.uk/about/teams/eukaryotic-annotation/>)
- 4) MAKER2 (<https://github.com/Yandell-Lab/maker>)
- 5) **Braker3** (<https://github.com/Gaius-Augustus/BRAKER>) -> Augustus
- 6) GeMoMa (<https://www.jstacs.de/index.php/GeMoMa>)
- 7) Deep learning annotation tools (Helixer, Tiberius)

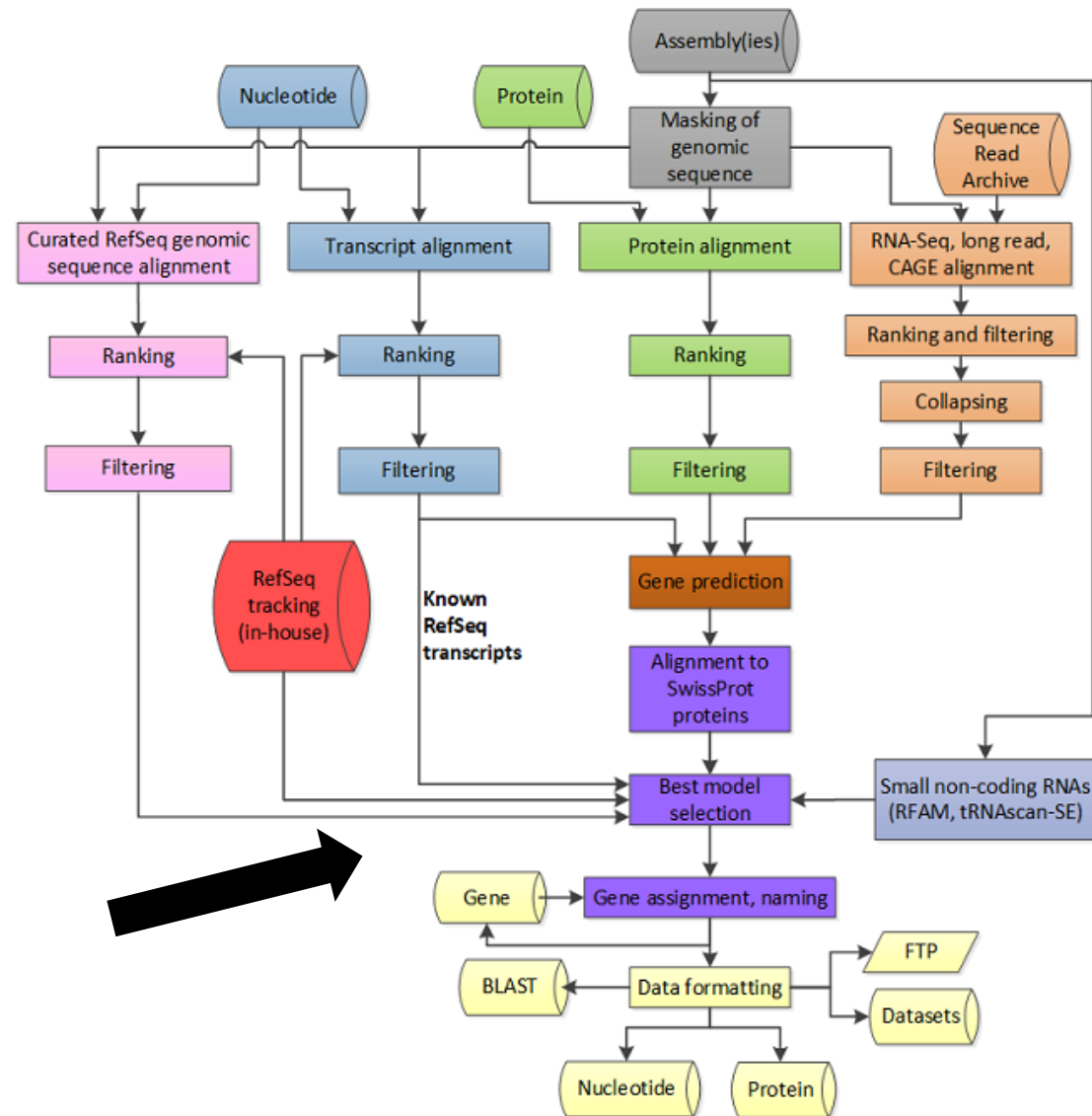
NCBI RefSeq-EGAP annotation workflow

- GenBank cDNAs and ESTs
- GenBank proteins
- RefSeq transcripts and proteins
- RNA-Seq (short-read data)
- PacBio IsoSeq, ONT transcriptomes
- CAGE and polyA-seq



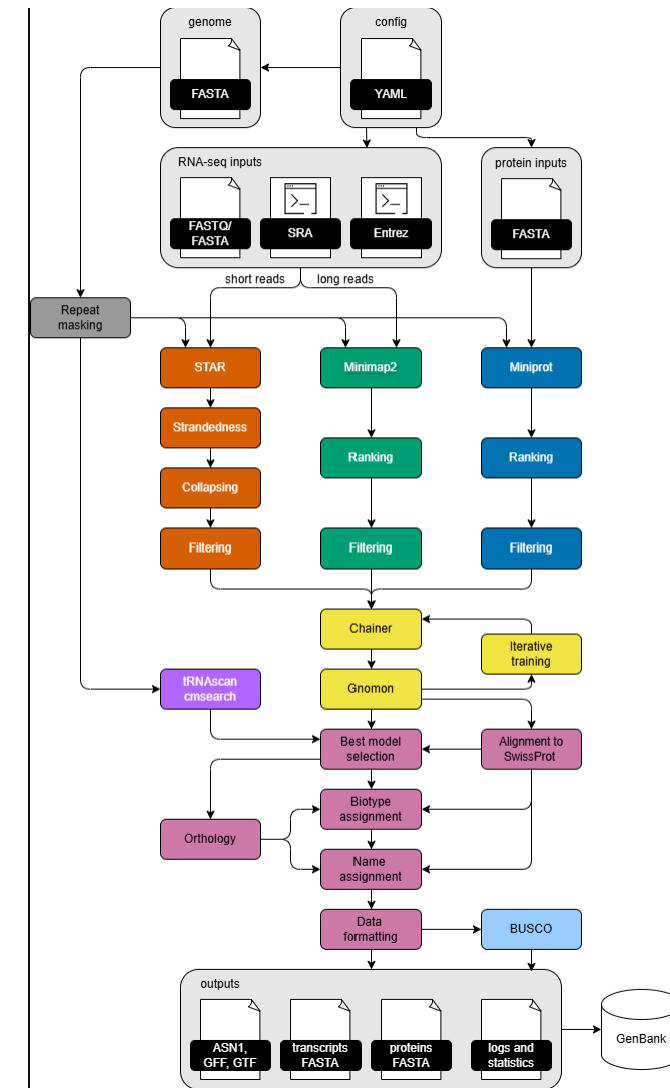
NCBI RefSeq-EGAP annotation workflow

- Orthology
- Assigning gene biotypes
- Nomenclature
- Gene expression



EGAPx annotation workflow

- Repeat masking
- RNAseq alignment performed through STAR aligner
- Miniprot/ProSplign for protein alignment
- Chainer combines partial alignments into longer models
- HMM-based program (GNOMON) processes the alignment data to generate final gene models
- GNOMON creates *ab initio* predictions where open reading frames of sufficient length but with no supporting alignment are detected
- tRNAscan and cmsearch for tRNA and non-coding-RNA identification
- Orthology and gene name assignment



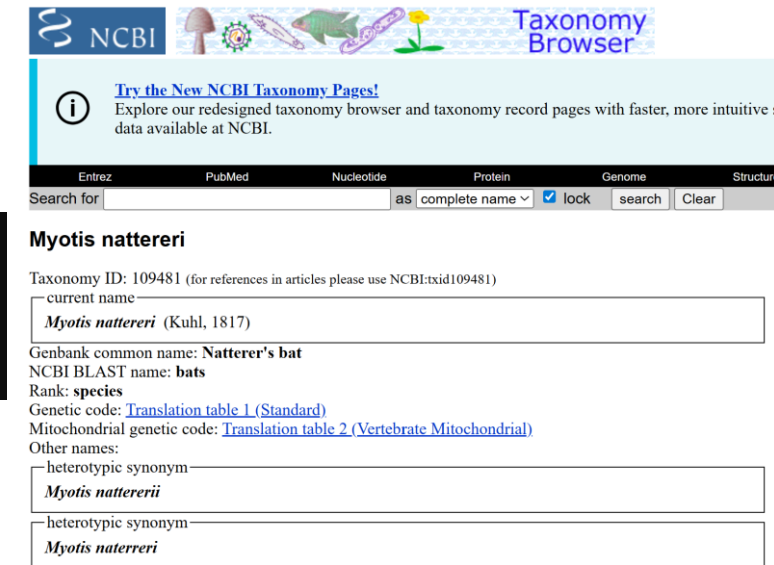
Input YAML

- Assembly FASTA • NCBI Taxid • RNA-seq data (more accurate annotations)
- Annotation provider name • Annotation name prefix • Locus_tag prefix
- Parameter can be changed from default

Output files

- Annotation: • GFF, GTF, ASN • FASTA: transcripts, CDS, proteins
- Logs: logs and reports (BUSCO, annotation metrics, presence of frameshifts and premature stop codons).
- Annotated feature stats

```
genome: https://ftp.ncbi.nlm.nih.gov/genomes/all/GCA/964/212/035/GCA_964212035.2_mMyoNat1.hap1.2/GCA_964212035.2_mMyoNat1.hap1.2_genomic.fna.gz
short_reads: txid109481[Organism] AND 75:350[ReadLength] AND biomol_rna[Properties] AND (illumina[Platform] OR dnbseq[Platform] OR bgiseq[Platform])
long_reads: txid109481[Organism] AND (oxford_nanopore[Platform] OR pacbio_smrt[Platform]) AND biomol_rna[Properties]
taxid: 109481
locus_tag_prefix: mMyoNat1_egapxtmp
genome_size_threshold: 400000000
tasks:
  star_index:
    STAR: --runThreadN 8 --limitGenomeGenerateRAM 50500500951
```



The screenshot shows the NCBI Taxonomy Browser interface. At the top, there's a navigation bar with icons for Entrez, PubMed, Nucleotide, Protein, Genome, and Structure. Below this is a search bar with the text "Search for" and a dropdown menu set to "complete name". To the right of the search bar are buttons for "lock", "search", and "Clear". The main content area displays the results for "Myotis nattereri". It includes the Taxonomy ID (109481), the current name "Myotis nattereri (Kuhl, 1817)", the Genbank common name "Natterer's bat", the NCBI BLAST name "bats", the Rank "species", the Genetic code "Translation table 1 (Standard)", and the Mitochondrial genetic code "Translation table 2 (Vertebrate Mitochondrial)". There are also sections for "Other names" and "heterotypic synonym" with input fields for "Myotis nattererii" and "Myotis nattereri".

Key factors for an effective gene model annotation

- 1) Good level of the assembly (Contiguity, Completeness)
- 2) Effective annotation of transposable elements
- 3) Valid *ab-initio* models
- 4) High variability of genes expressed (RNA-seq datasets deriving from different tissues, developmental stages and/or ages)
- 5) Use of curated proteomes

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Quality control

❖ Metrics

Gene count, transcript count, mono-exonic to multi-exonic gene ratio, median exons per transcript, maximum exons per transcript, and median transcript length.

❖ BUSCO and OMArk

Evaluate genome annotation completeness based on conserved orthologs, OMArk provides some additional details even on the consistency (presence or absence evaluation but does not ensure correct gene structure).

❖ Accuracy metrics

Sensitivity (or Recall) measures how many features in the reference are found in the query

Specificity (or Precision) measures how many features in the query match the reference

F1-score = $(2 \times \text{Recall} \times \text{Precision}) / (\text{Recall} + \text{Precision})$

AED score -> exon-level coverage and identity between the annotation and evidence

Psauron -> support to identify potentially incorrect annotations

scovhsp and qcovhsp -> measure the coverage of alignments

❖ Visualize tracks on IGV/Jbrowse2

❖ Metrics

Gene count, transcript count, mono-exonic to multi-exonic gene ratio, median exons per transcript, maximum exons per transcript, and median transcript length.

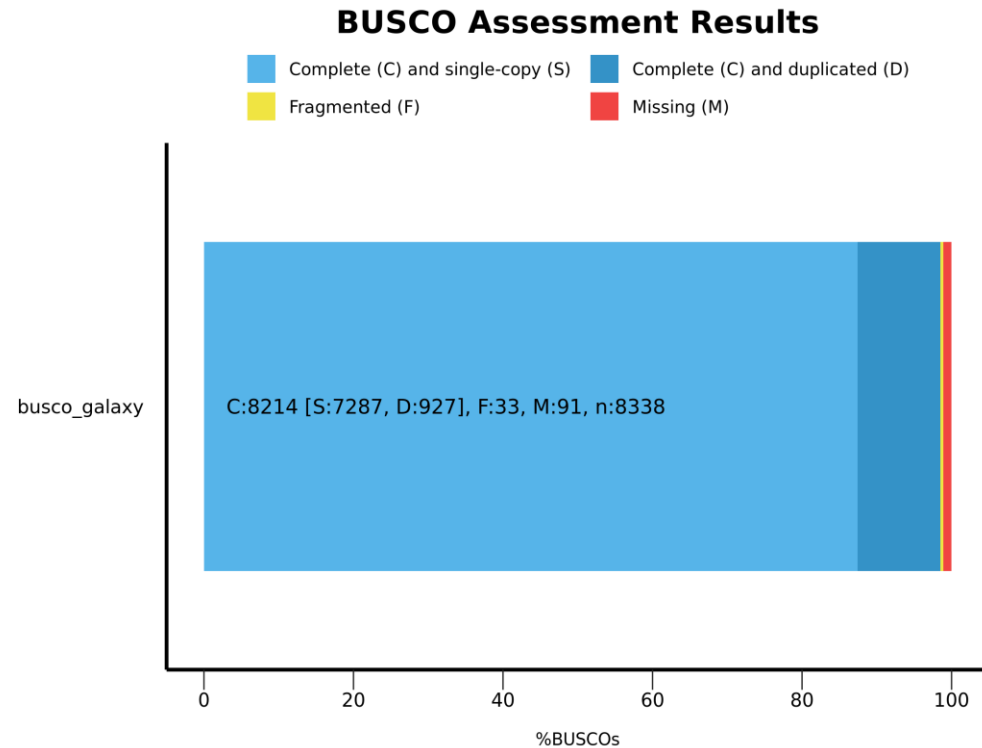
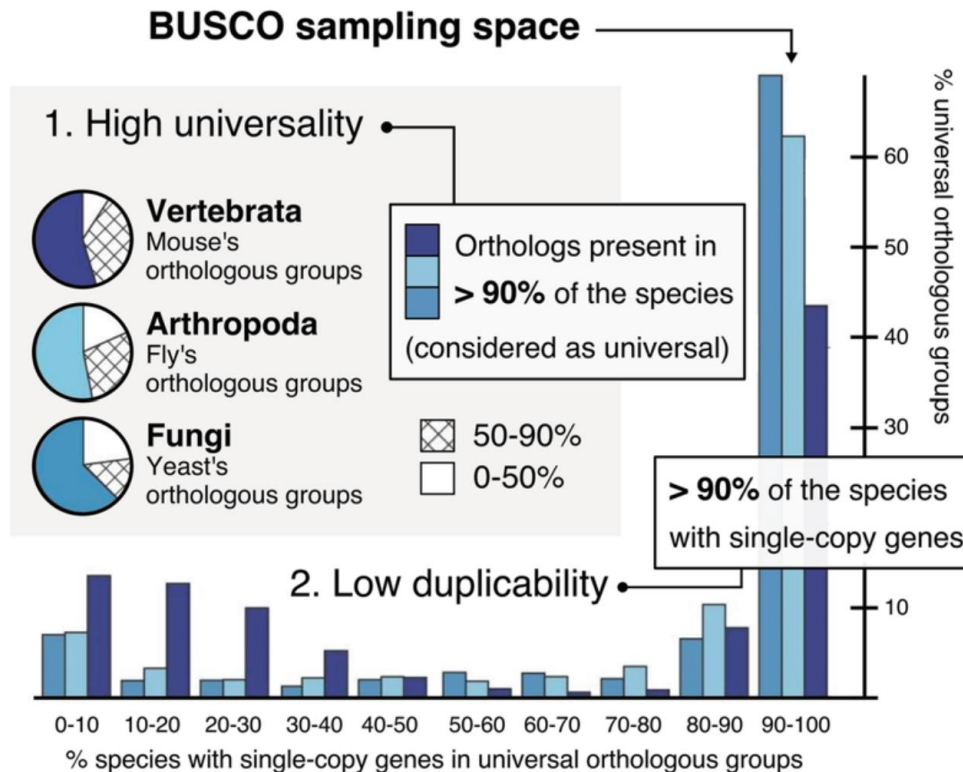
Number of gene	1214
Number of transcript	1708
Number of mrnas with utr both sides	1049
Number of mrnas with at least one utr	1235
Number of cds	1643
Number of exon	19831
Number of five_prime_utr	1150
Number of start_codon	1636
Number of stop_codon	1643
Number of three_prime_utr	1134
Number of exon in cds	18711
Number of exon in five_prime_utr	1774
Number of exon in three_prime_utr	1192
Number of intron in cds	17068
Number of intron in exon	18123
Number of intron in five_prime_utr	624
Number of intron in start_codon	1
Number of intron in stop_codon	3
Number of intron in three_prime_utr	58
Number gene overlapping	14
Number of single exon gene	43
Number of single exon transcript	43
mean transcripts per gene	1.4
mean cdss per transcript	1.0
mean exons per transcript	11.6
mean five_prime_utrs per transcript	0.7
mean start_codons per transcript	1.0
mean stop_codons per transcript	1.0
mean three_prime_utrs per transcript	0.7
mean exons per cds	11.4
mean exons per five_prime_utr	1.5
mean exons per three_prime_utr	1.1
mean introns in cdss per transcript	10.0

```
<OverallCounts>
  <Counts feature_type="genes">
    <Total>27457</Total>
    <Subcount subtype="other">30</Subcount>
    <Subcount subtype="non-transcribed pseudo">947</Subcount>
    <Subcount subtype="transcribed pseudo">2</Subcount>
    <Subcount subtype="has variants">11276</Subcount>
    <Subcount subtype="partial">167</Subcount>
    <Subcount subtype="Ig TCR segment">20</Subcount>
    <Subcount subtype="non coding">8074</Subcount>
    <Subcount subtype="protein coding">18384</Subcount>
    <Subcount subtype="major correction">570</Subcount>
    <Subcount subtype="minor correction">0</Subcount>
    <Subcount subtype="premature stop">54</Subcount>
    <Subcount subtype="has frameshifts">543</Subcount>
```

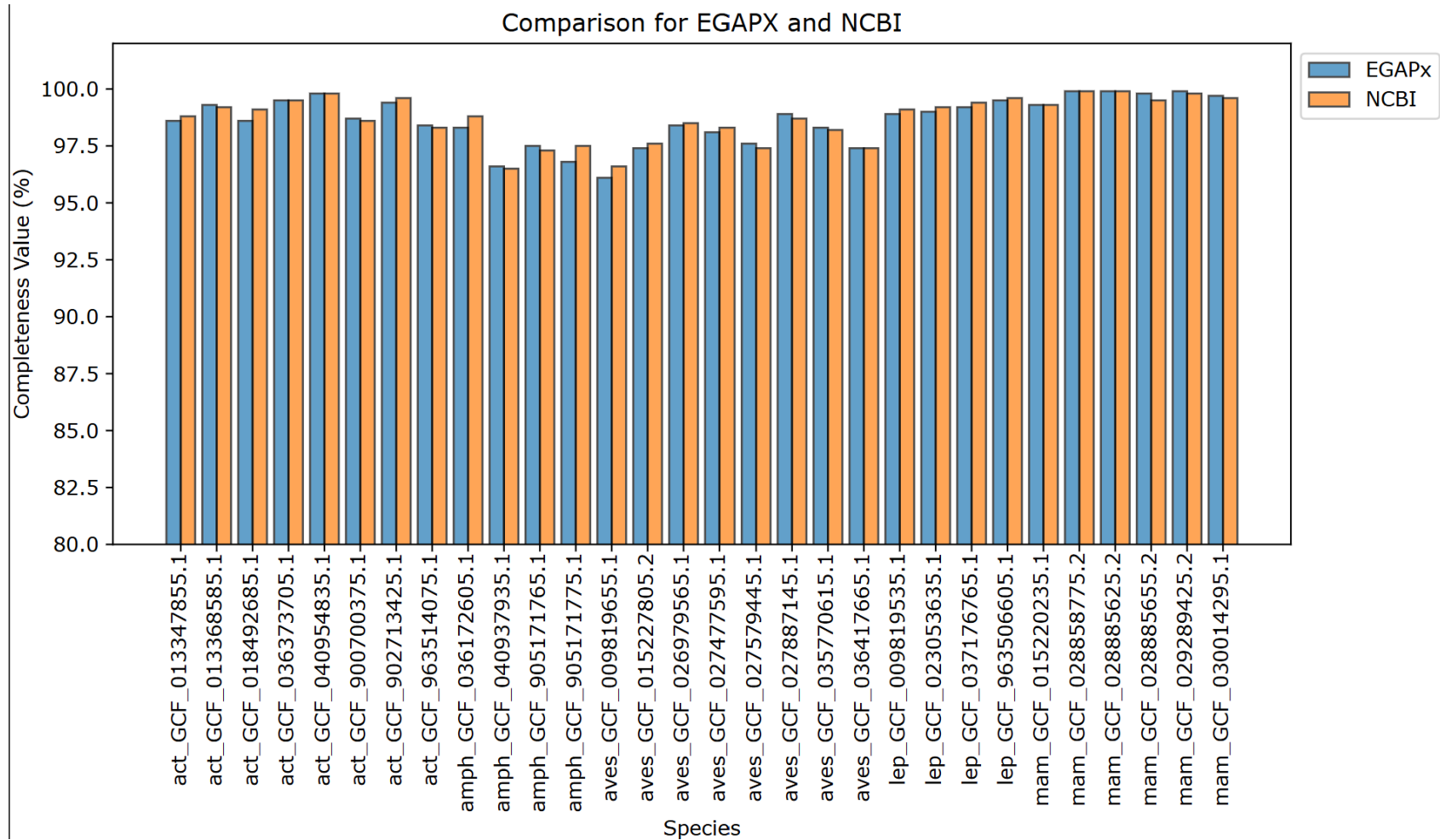


BUSCO (Benchmarking Universal Single-Copy Orthologs)

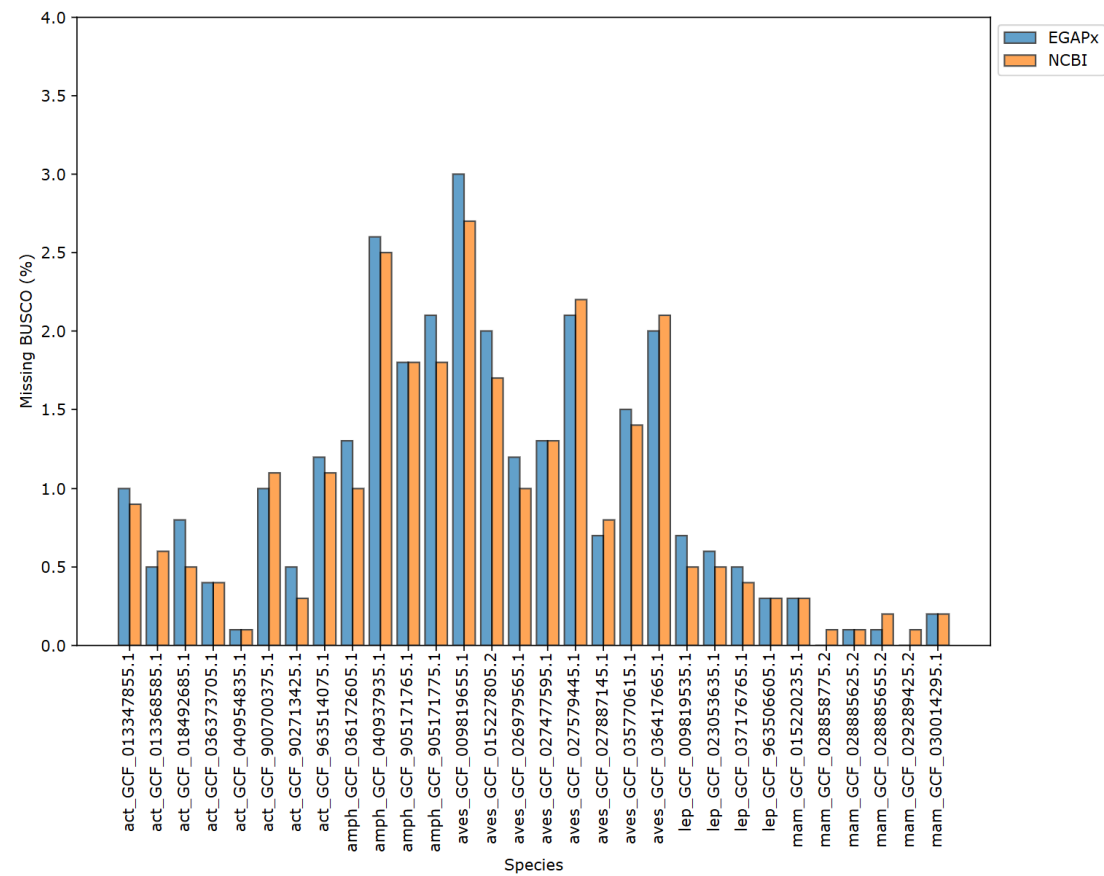
- <https://busco.ezlab.org/> (Galaxy, Conda package and Docker container)
- Detects sensitivity in lineage-specific conserved genes



Example of a BUSCO Completeness result on some VGP genomes

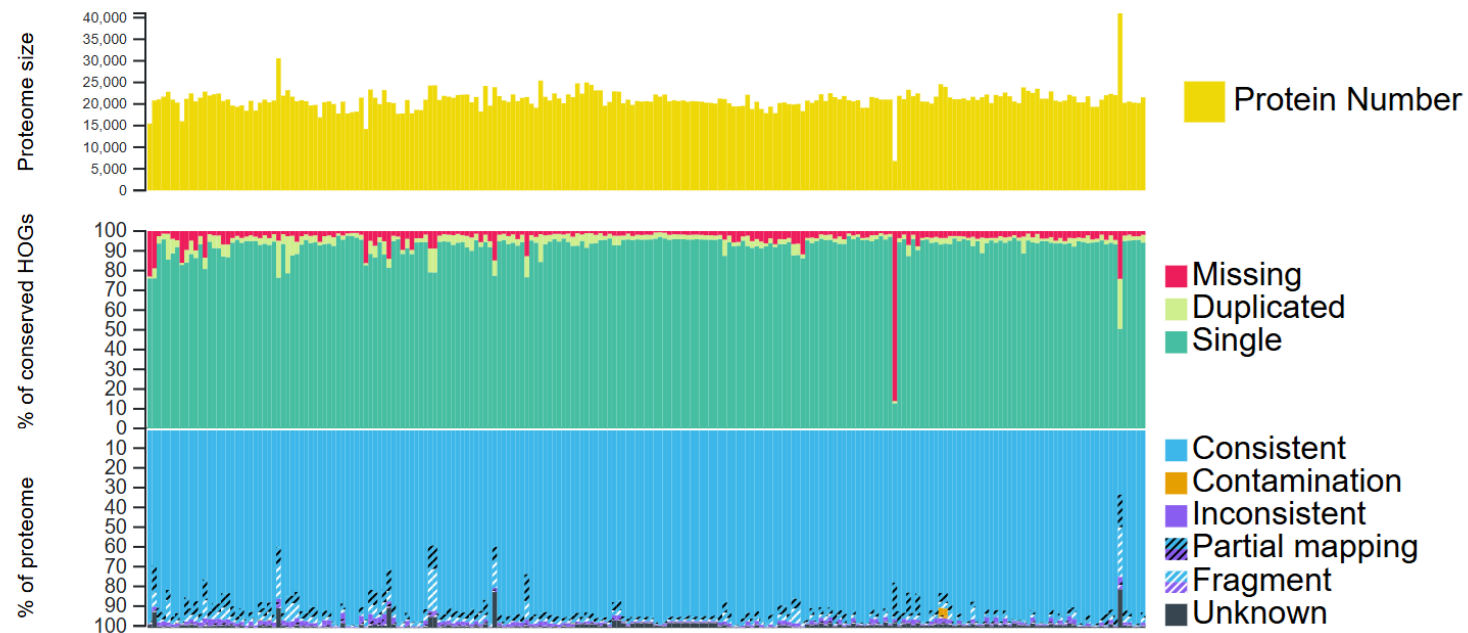


Example of a BUSCO result

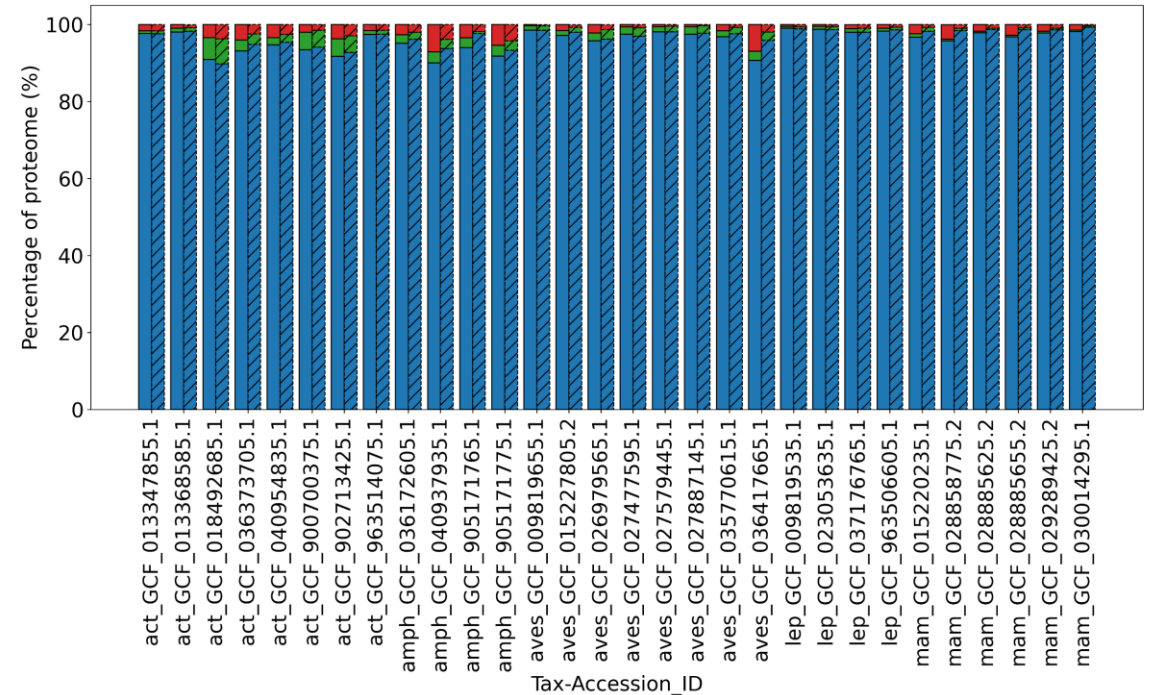
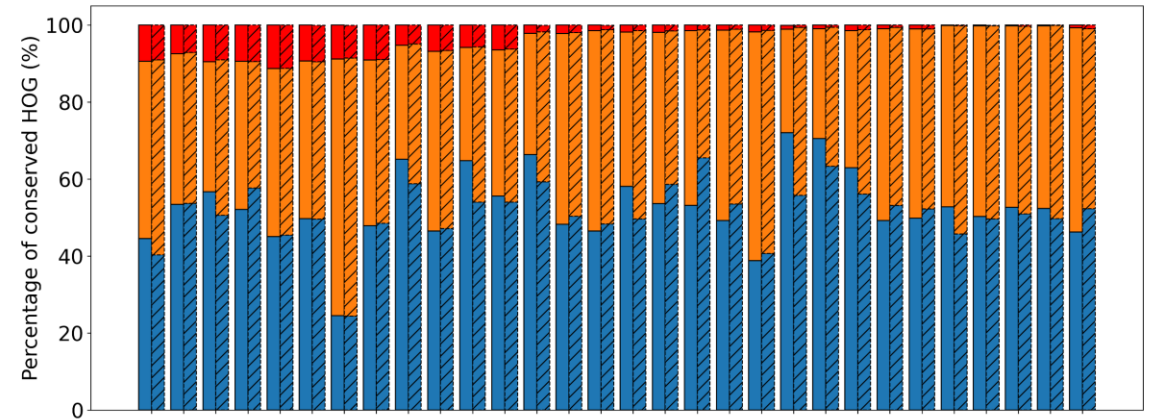


OMArk

- <https://omark.omabrowser.org/> - <https://github.com/DessimozLab/OMArk> (Galaxy, Conda Package)
- It is recommended that the database used as reference from OMArk covers a wide range of species. Authors suggest to use constructed from the whole OMA database (LUCA.h5)
- OMArk: detects contamination, sensitivity, etc...



OMark (example of EGAPx vs RefSeq)



OMArk (contamination case)

Ochotona sp. (pika)

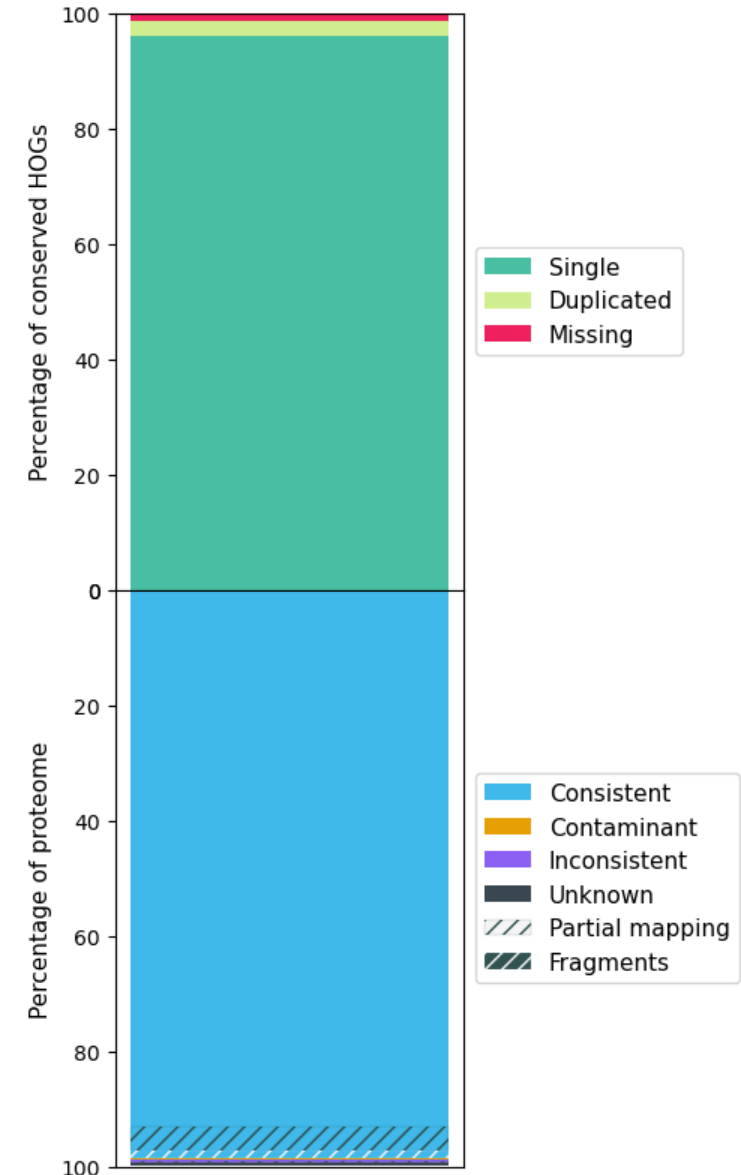
#Potential contaminant N°1

Clade: Eimeriorina (Apicomplexa)

Number of associated query proteins: 66 (0.32%)



These proteins map to unplaced scaffolds



PSAURON

JOURNAL ARTICLE EDITOR'S CHOICE

PSAURON: a tool for assessing protein annotation across a broad range of species

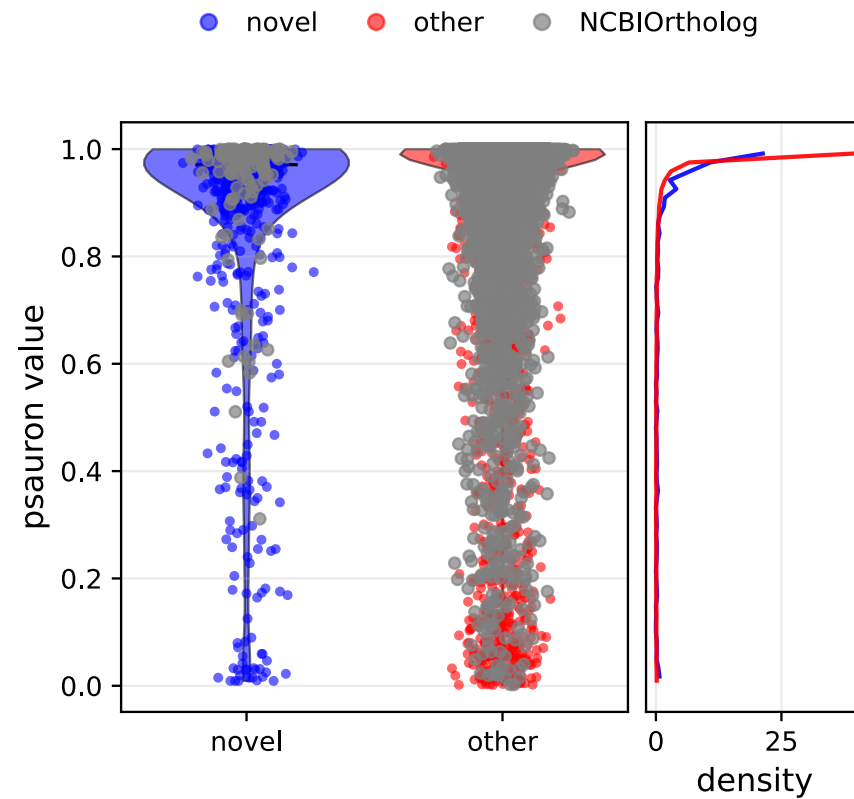
Markus J Sommer , Aleksey V Zimin, Steven L Salzberg

NAR Genomics and Bioinformatics, Volume 7, Issue 1, March 2025, lqae189,
<https://doi.org/10.1093/nargab/lqae189>

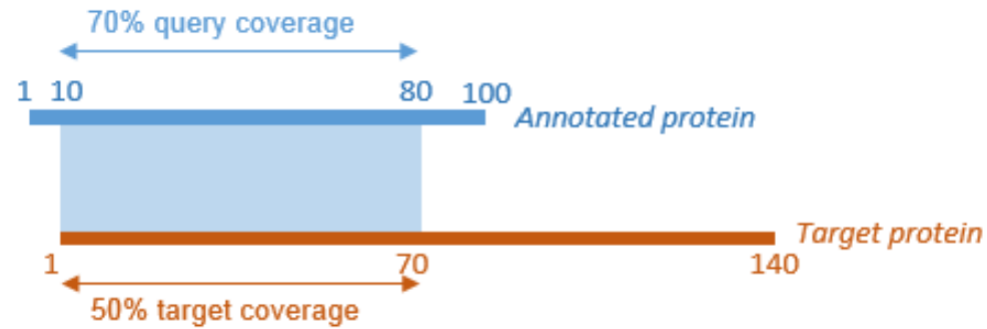
Published: 07 January 2025 **Article history** ▼

- Uses a **Temporal Convolutional Network (TCN)** machine learning model to assess whether sequences are likely to represent true protein-coding genes.
- Model trained on a **large, diverse dataset of plant and animal proteomes** (1000 species), allowing it to recognize general coding patterns **without species-specific retraining**.
- Support to identify **potentially incorrect annotations** that require further investigation
- Fast, minimal computational requirements

PSAURON

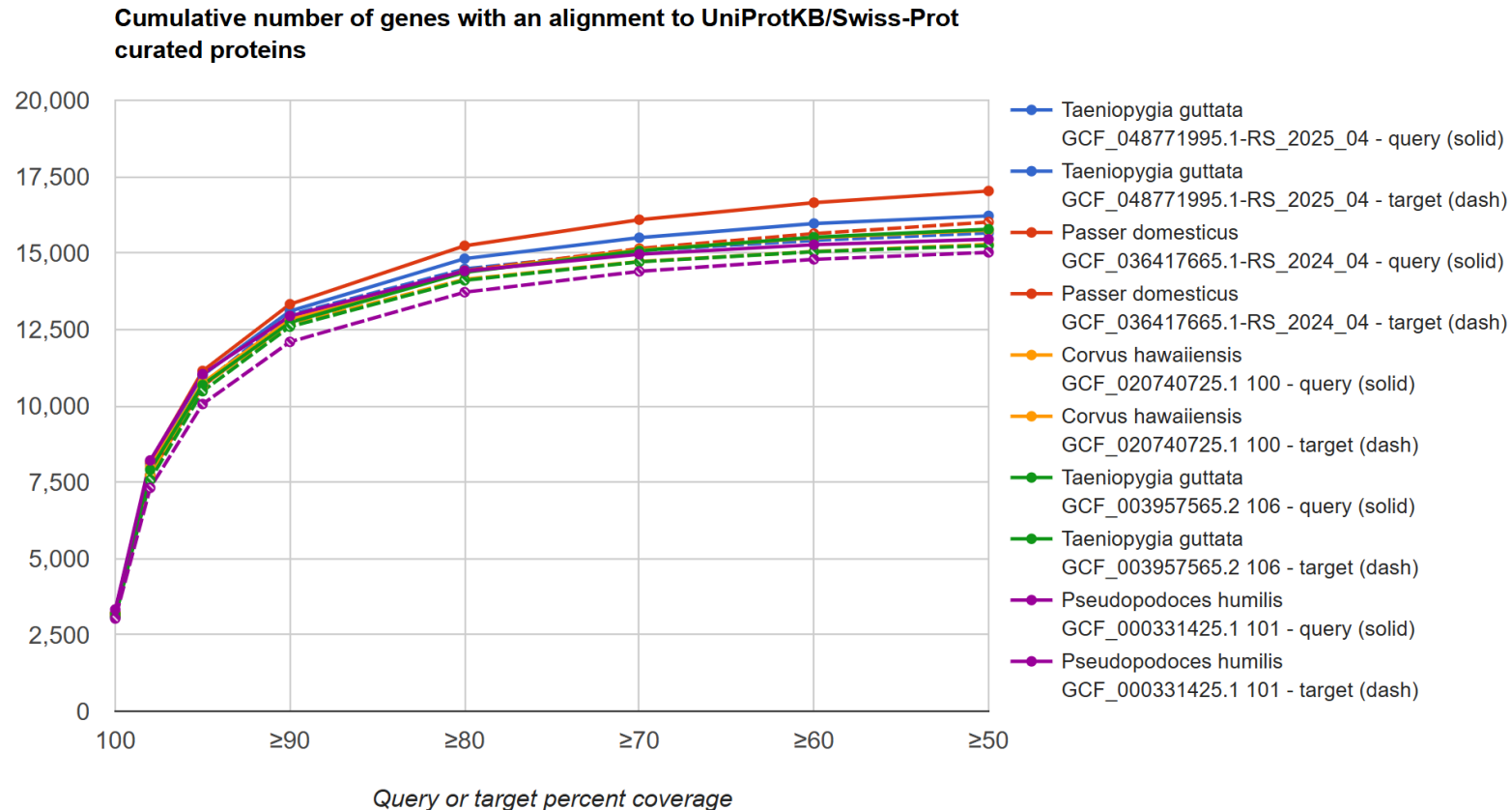


scovhsp and qcovhsp



The query coverage is the percentage of the annotated protein length that is included in the alignment. The target coverage is the percentage of the target length that is included in the alignment.

scovhsp and qcovhsp



The final set of annotated proteins was searched with diamond BLASTP against the UniProtKB/Swiss-Prot curated proteins, using the annotated proteins as the query and the high-quality proteins as the target.

Accuracy metrics

<https://github.com/gpertea/gffcompare>

#-----	Sensitivity	Precision	
Base level:	60.3	78.9	
Exon level:	70.8	70.7	
Intron level:	80.2	81.0	

Missed exons:	44906/356755	(12.6%)
Novel exons:	55101/340400	(16.2%)
Missed introns:	16895/295269	(5.7%)
Novel introns:	31851/292069	(10.9%)
Missed loci:	8444/40672	(20.8%)
Novel loci:	14047/47382	(29.6%)

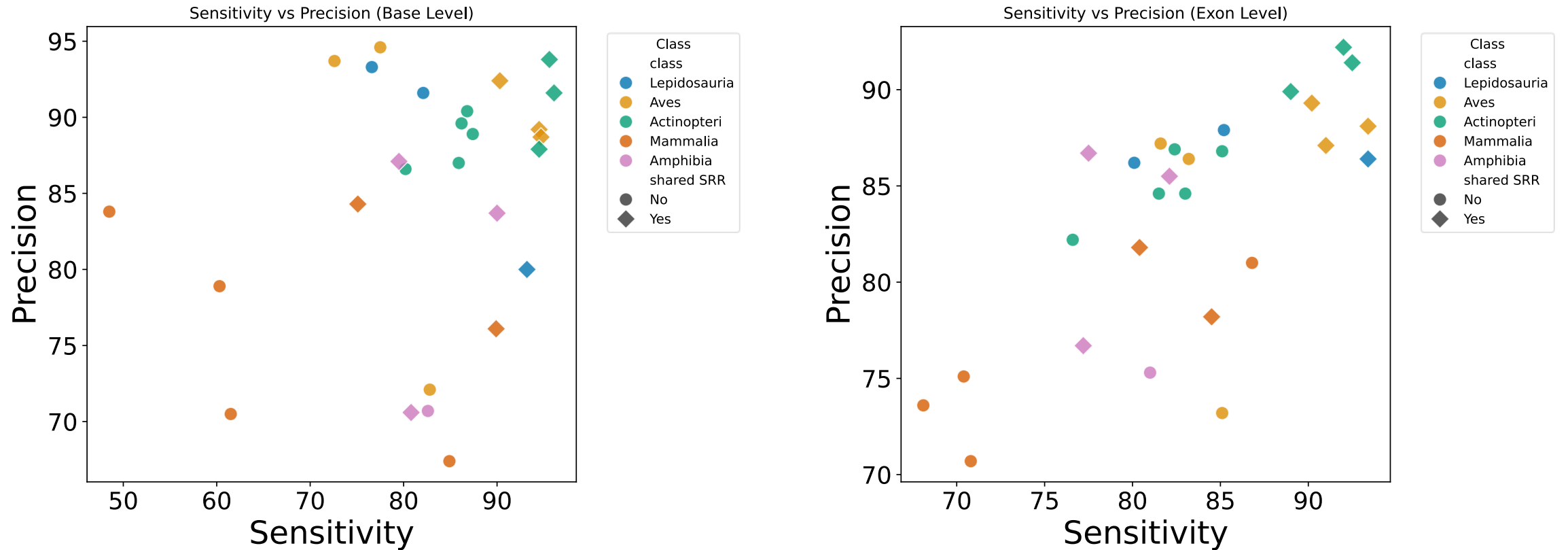
Sensitivity (or Recall) measures how many features in the reference (RefSeq NCBI) are found in the query (EGAPx).

Specificity (or Precision) measures how many features in the query (EGAPx) match the reference.

$$\mathbf{F1\text{-}score} = (2 \times \mathbf{Recall} \times \mathbf{Precision}) / (\mathbf{Recall} + \mathbf{Precision})$$

Accuracy metrics

<https://github.com/gpertea/gffcompare>

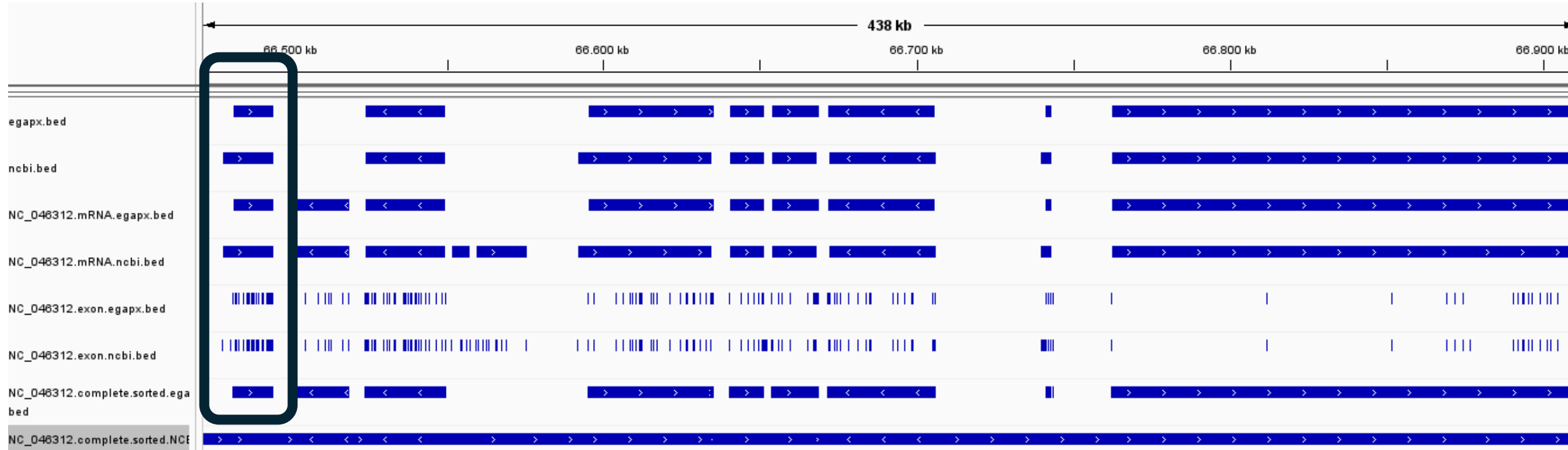


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IGV: Visualization tool



GENE ID: EIF4G2: eukaryotic translation initiation factor 4 gamma 2
Different exon recognition

Outline of the seminar

- QC on Genome Assembly
- Overview of TE annotation
- Introduction on the annotation process
- Pipelines for the genome annotation
- Focus on EGAP/EGAPx Pipeline
- Evaluation of the gene annotation (QC)
- **Functional annotation**
- Summary

What is Gene Functional Annotation?

- Assign biological meaning to identified gene models
- Sequence Homology and Functional homology
- **Challenges**
 1. many “hypothetical proteins” remain uncharacterized
 2. large gene families with many paralogues (i.e. keratin gene family)
- **Principle of “Guilty by association”** -> expression, protein–protein interactions (as a complementary approach)

Sequence Homology vs Functional homology based tools

- BLAST / DIAMOND
- **Orthology**: eggNOG-mapper, OrthoDB, OrthoFinder (Reciprocal Best Hit).
- HMMER / Pfam – domain detection via profile HMMs;
- InterProScan aggregates signatures.

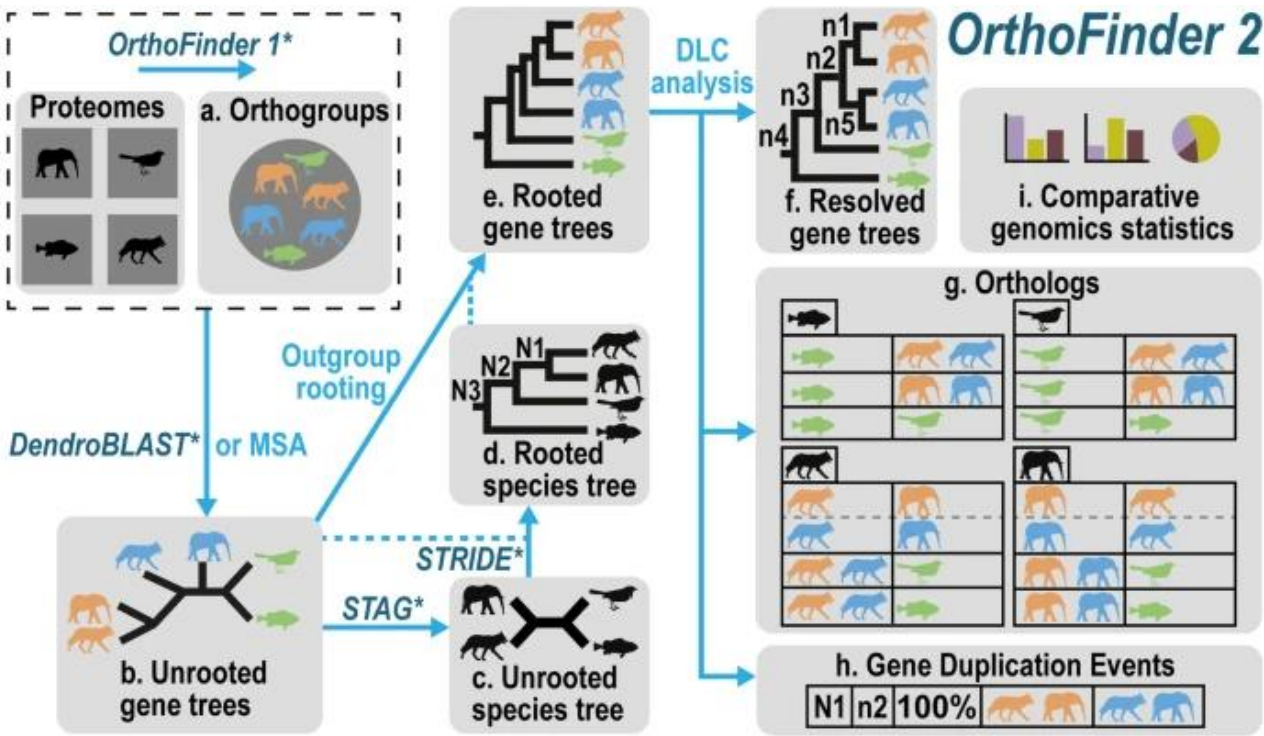
Orthology based: Orthofinder

Software | [Open access](#) | Published: 14 November 2019

OrthoFinder: phylogenetic orthology inference for comparative genomics

[David M. Emms](#) & [Steven Kelly](#) 

Genome Biology **20**, Article number: 238 (2019) | [Cite this article](#)



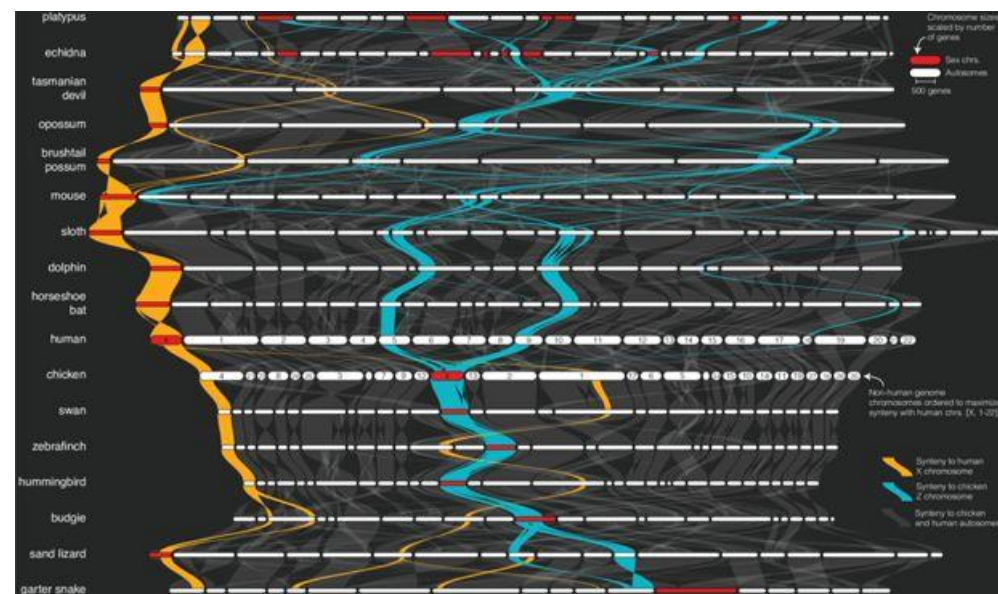
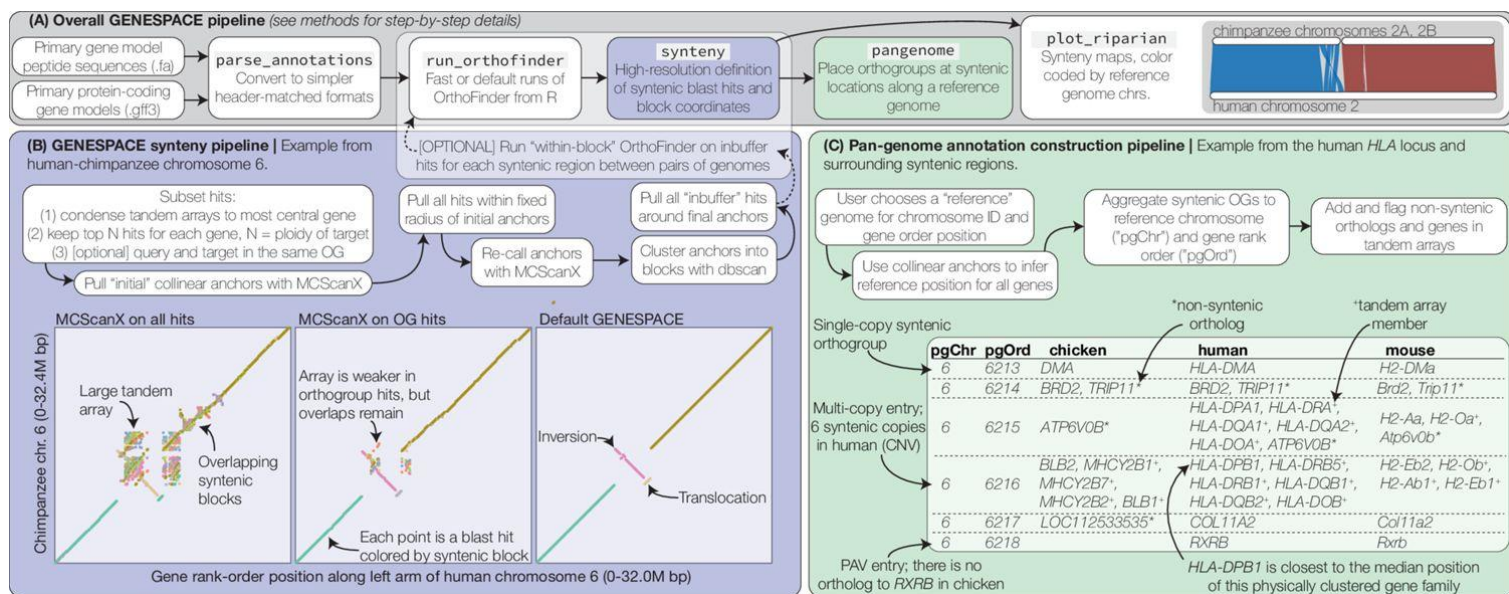
Orthogroup	Species1	Species2	Species3	Species4	Species5	Species6	Species7	Species8	Species9
OG0000000	329	346	970	38	56	226	331	575	22
OG0000001	361	19	217	8	12	33	220	225	4
OG0000002	77	17	59	16	24	93	52	94	185
OG0000003	46	41	154	0	7	44	81	98	18
OG0000004	47	20	65	15	17	99	51	59	89
OG0000005	8	4	5	83	43	74	3	6	0
OG0000006	27	39	14	2	4	9	16	17	158
OG0000007	40	36	15	23	10	15	18	18	35
OG0000008	5	142	6	6	4	6	6	5	115
OG0000009	0	0	0	54	68	0	0	0	0
OG0000010	30	33	33	8	13	25	28	26	42
OG0000011	13	42	12	4	3	8	8	11	3
OG0000012	55	1	56	0	1	39	45	42	0
OG0000013	29	4	24	12	14	12	23	23	40
OG0000014	31	46	27	13	12	32	26	33	15
OG0000015	23	8	36	9	9	28	25	23	19
OG0000016	11	7	32	5	11	19	11	11	21
OG0000017	22	12	24	14	16	20	28	30	17
OG0000018	23	19	27	13	12	26	23	22	18

Orthogroup: a set of genes across multiple genomes derived from a single ancestral gene.

Paralog: orthogroup members derived from a duplication event since speciation.

GENESPACE

A tool that integrates sequence similarity with synteny, or conserved gene order, to improve cross-species genome comparisons, helping researchers detect gene loss and duplication over long evolutionary timescales



Functional annotation: InterProScan

- Combine different tools, it can annotate PFAM domains and classify proteins based on Gene Ontologies

Ontologies (species-independent manner) define gene products based on their:

- **Biological processes:** series of molecular events (i.e., apoptosis)
- **Cellular components:** Locations relative to cellular structures in which a gene product performs a function (i.e. extracellular matrix)
- **Molecular functions:** activities of a gene product at the molecular level, such as binding, polymerase, catalysis and so on.

Functional annotation

- **InterProScan** for protein domain homology (<https://github.com/ebi-pf-team/interproscan>)



PROSITE is a database of protein families and domains. It consists of biologically significant sites, patterns and profiles that help to reliably identify to which known protein family a new sequence belongs. PROSITE is based at the Swiss Institute of Bioinformatics (SIB), Geneva, Switzerland.



HAMAP stands for High-quality Automated and Manual Annotation of Proteins. HAMAP profiles are manually created by expert curators. They identify proteins that are part of well-conserved protein families or subfamilies. HAMAP is based at the SIB Swiss Institute of Bioinformatics, Geneva, Switzerland.



Pfam is a large collection of multiple sequence alignments and hidden Markov models covering many common protein domains. Pfam is based at the Wellcome Trust Sanger Institute, Hinxton, UK.



PRINTS is a compendium of protein fingerprints. A fingerprint is a group of conserved motifs used to characterise a protein family or domain. PRINTS is based at the University of Manchester, UK.



ProDom protein domain database consists of an automatic compilation of homologous domains. Current versions of ProDom are built using a novel procedure based on recursive PSI-BLAST searches. ProDom is based at PRABI Villeurbanne, France.



SMART (a Simple Modular Architecture Research Tool) allows the identification and annotation of genetically mobile domains and the analysis of domain architectures. SMART is based at EMBL, Heidelberg, Germany.



TIGRFAMs is a collection of protein families, featuring curated multiple sequence alignments, hidden Markov models (HMMs) and annotation, which provides a tool for identifying functionally related proteins based on sequence homology. TIGRFAMs is based at the J. Craig Venter Institute, Rockville, MD, US.



PIRSF protein classification system is a network with multiple levels of sequence diversity from superfamilies to subfamilies that reflects the evolutionary relationship of full-length proteins and domains. PIRSF is based at the Protein Information Resource, Georgetown University Medical Centre, Washington DC, US.



SUPERFAMILY is a library of profile hidden Markov models that represent all proteins of known structure. The library is based on the SCOP classification of proteins: each model corresponds to a SCOP domain and aims to represent the entire SCOP superfamily that the domain belongs to. SUPERFAMILY is based at the University of Bristol, UK.



CATH-Gene3D database describes protein families and domain architectures in complete genomes. Protein families are formed using a Markov clustering algorithm, followed by multi-linkage clustering according to sequence identity. Mapping of predicted structure and sequence domains is undertaken using hidden Markov models libraries representing CATH and Pfam domains. CATH-Gene3D is based at University College, London, UK.



PANTHER is a large collection of protein families that have been subdivided into functionally related subfamilies, using human expertise. These subfamilies model the divergence of specific functions within protein families, allowing more accurate association with function, as well as inference of amino acids important for functional specificity. Hidden Markov models (HMMs) are built for each family and subfamily for classifying additional protein sequences. PANTHER is based at University of Southern California, CA, US.



Entry matches to this protein¹

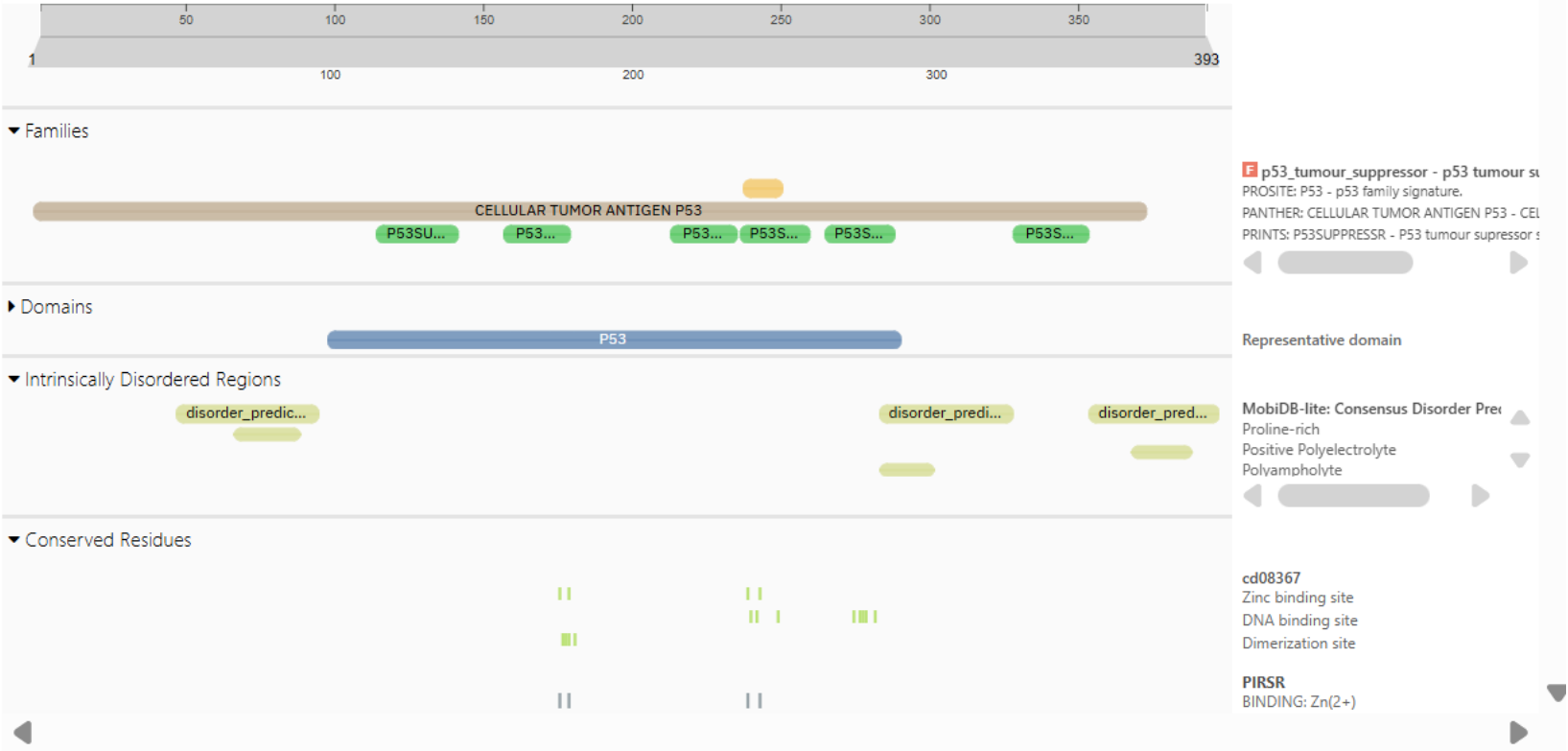
Options

Download

Feature Display Mode ?

Summary

Full



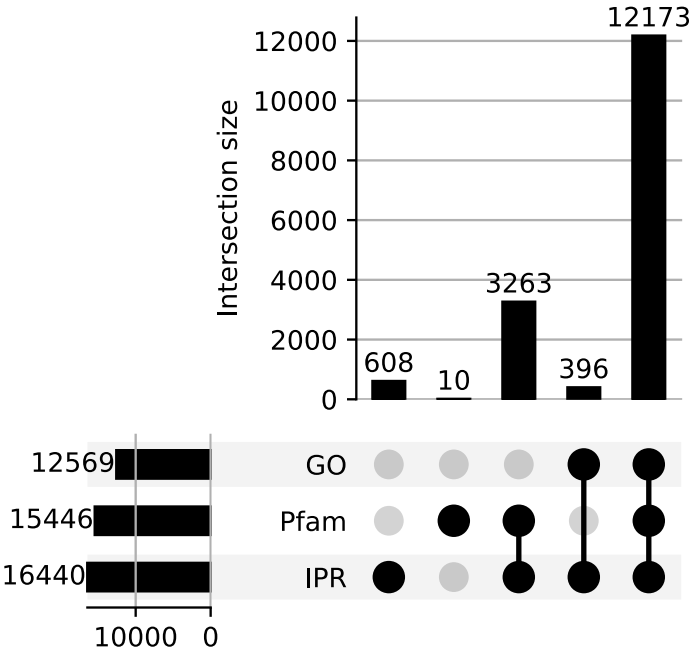
InterPro GO terms

Biological Process	Molecular Function	Cellular Component
- regulation of DNA-templated transcription (GO:0006355) ↗ - apoptotic process (GO:0006915) ↗ - protein tetramerization (GO:0051262) ↗	- DNA binding (GO:0003677) ↗ - DNA-binding transcription factor activity (GO:0003700) ↗ - transcription cis-regulatory region binding (GO:0000976) ↗ - protein binding (GO:0005515) ↗	nucleus (GO:0005634) ↗

PANTHER GO terms

Biological Process	Molecular Function	Cellular Component
regulation of transcription by RNA polymerase II (GO:0006357) ↗	- DNA-binding transcription factor activity, RNA polymerase II-specific (GO:0000981) ↗ - RNA polymerase II cis-regulatory region sequence-specific DNA binding (GO:0000978) ↗	None

Ex. InterProScan results



Functional annotation: Protein structure prediction

- Development of **AlphaFold** and **RoseTTAFold** to predict 3D protein structures directly from amino acid sequences, even without clear sequence homology.
- Structure prediction also provides an effective method for distinguishing between functional and non-functional isoforms of protein-coding genes (QC)
- Deep learning methods to predict interactions between proteins, and between proteins and nucleic acids, potentially broadening the utility of structure-based functional prediction.

Principle of “Guilty by association”

- Co-expression analysis, protein–protein interactions (Weighted Gene Co-expression Network Analysis)
- Complements homology-based annotation, especially for fast-evolving or novel genes
- High-throughput loss-of-function experiments (functional genomics approaches), such as CRISPR–Cas9 and RNAi-based screening, can empirically link genes to phenotypes, providing valuable insight into their biological roles.

Take home message

- Perform a QC at each step of your workflow
- Annotation in eukaryotes is still a challenging process
- Support from HMM models + RNA-seq and protein evidences
- Popular annotation tools/pipelines:
 1. EGAPx, BRAKER3
 2. Tools exploiting deep learning (i.e. Tiberius, Helixer)
- QC (BUSCO, OMArk, general metrics, comparative analysis, comparison vs a reference)
- Good BUSCO/OMArk results doesn't mean a good annotation
- Check for sequence and functional annotation
- Comparative analysis aids in the interpretation of results

Thanks for the attention!!

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https://github.com/marco91sol/SequAna_course

• GFF3 Format (Version 3)

Tab-delimited text file for describing sequence features

- Supports annotations such as:

- CDS, exons, 5'UTR, 3' UTR, microRNAs, binding domains, etc

1) **seqid** – sequence identifier ; 2) **source** – the annotation source ; 3) **type** – the feature type ; 4–5. **start** and **end** – feature coordinates ; 6) **score** – numerical score (if available) ; 7) **strand** – strand information (+/–) ; 8) **phase** – reading frame;

9) **attributes** – additional feature details

Seqid	Source	Type	Start	End	Score	Strand	Phase	Attributes
##gff-version 3								
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Chr1	phytozomev12	mRNA	3631	5899	.	+	.	ID=AT1G01010.1.Araport11.447;Parent=AT1G01010.Araport11.447;Name=AT1G01010.1;geneName=NAC001;longest=1;pacid=37401853
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Chr1	phytozomev12	exon	8417	8464	.	-	.	ID=AT1G01020.1.Araport11.447.exon.2;Parent=AT1G01020.1.Araport11.447;pacid=37399351
Chr1	phytozomev12	CDS	8417	8464	.	-	0	ID=AT1G01020.1.Araport11.447.CDS.2;Parent=AT1G01020.1.Araport11.447;pacid=37399351